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Microscopic rate coefficients in reactions with flexible transition states: Analysis of the transitional-mode sum of states

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Expressions are derived for the energy- and angular-momentum-resolved transitional-mode sum of states, $W_{\text{TM}}(E, J)$, for flexible transition states in unimolecular, recombination, or bimolecular collision-complex-forming reactions. The expressions are derived classically by evaluation of the phase-space volume integral. The phase-space integral is so arranged that the total available momentum-space volume, $\Phi(E, J, \mathbf{q})$, for a frozen configuration $\mathbf{q}$ is first evaluated. Accurate analytic expressions for $\Phi(E, J, \mathbf{q})$ are obtained for all relevant pairwise combinations of atom, linear, spherical-top, symmetric-top, and asymmetric-top fragments in flexible transition states. The analytic expressions for $\Phi(E, J, \mathbf{q})$ indicate clearly the conditions under which the common method of imposing angular momentum conservation, which assumes that $J \approx L$ (L being the orbital angular momentum), will fail. $W_{\text{TM}}(E, J)$ is then obtained by integration of $\Phi(E, J, \mathbf{q})$ over configuration space. Exact evaluation of the integral over configuration space involves at most a five-dimensional numerical integral. Accurate analytical expressions for $W_{\text{TM}}(E, J)$ are derived for model potentials which admit sufficient flexibility for the fitting of more-accurate potentials. These expressions enable the calculation of accurate microscopic rate coefficients $k(E, J)$ by microcanonical variational Rice-Ramsperger-Kassel-Marcus (RRKM) theory at little more computational expense than a standard RRKM calculation.

I. INTRODUCTION

Chemical reactions involving a potential surface with no barrier to recombination are ubiquitous in gas-phase chemistry; occurring commonly in simple-fission unimolecular dissociation, radical-radical recombination, ion-molecule reactions and the reverse decomposition of molecular ions, and collision-complex-forming bimolecular reactions. The accurate kinetic modeling of these reactions is a complex and difficult task. Currently, the only viable prospect for modeling and predicting such rate coefficients for a wide range of reactive systems of differing complexity is via statistical theory, with the use of either the variational (e.g., Refs. 1 and 2) or adiabatic (e.g., Ref. 3) formalisms. It is well known that angular momentum plays a crucial role in the dynamics of such reactions (e.g., Ref. 4). This has important consequences in the prediction and modeling of rate coefficients both at high pressures,^{5,2} and in the falloff regime.⁶⁻⁸ In addition, angular momentum can play a crucial role in determining branching ratios for different channels in bimolecular collision-complex-forming reactions⁹ or multichannel unimolecular reactions.¹⁰ Detailed experimental data are now available for certain reactions which present a sound basis against which theories can be compared, both on a thermal level (e.g., Ref. 11) and on a microscopic level (e.g., Refs. 12-15).

The calculation of microscopic rate coefficients $k(E, J)$ for a specified total energy E and angular momentum J is necessary for the solution of the master equation in predicting or modeling falloff effects (e.g., Ref. 16) and also for detailed comparison between theories or between theory and

experiment. The statistical expression for the unimolecular microscopic rate coefficient is

$$k(E, J) = W(E, J) / h \rho(E, J), \quad (1)$$

where $W(E, J)$ is the number of open channels between reactant and products (to be specified according to the branch of statistical theory invoked), h is Planck's constant, and $\rho(E, J)$ is the reactant density of states.

One may identify three principal difficulties associated with the calculation of $k(E, J)$ for systems without a barrier to recombination: the location of the transition state, angular momentum conservation, and, finally, accounting for the loose, anharmonic interaction potential between the fragments. These problems, and theory previously developed to solve them, are summarized below.

In the absence of any pronounced chemical barrier, there is no obvious position for the location of the transition state. Two alternative theories have been applied in the resolution of this problem: variational Rice-Ramsperger-Kassel-Marcus theory (VRRKM), which combines variational transition-state theory with the more traditional RRKM theory for unimolecular reactions, and the statistical adiabatic channel model (SACM). Microcanonical variational transition-state theory (e.g., Refs. 17-19 and 1) arises from the fact that a rate coefficient calculated with the use of the transition-state assumption must, in the absence of tunneling effects, be an upper bound to the exact rate coefficient (see, e.g., Ref. 16). The transition state is therefore taken to lie at that position along the reaction coordinate for which the sum of states $W(E, J)$ for the modes orthogonal to the reaction coordinate is a minimum, since this is the best ap-

proximation one can make to the true rate coefficient within the framework of transition-state theory. The SACM (e.g., Refs. 3 and 20) applies an adiabatic assumption to the rovibrational eigenstates of the degrees of freedom orthogonal to the reaction coordinate. The unimolecular reactant (or, collision complex, for the reverse recombination process) is assumed, as in other statistical theories, to behave ergodically on the time scale of reaction. However, it is assumed that the rate-determining dynamics occurs at separations of the fragments such that the quantum number labeling the rovibrational eigenstate for the modes orthogonal to the reaction coordinate is a conserved quantity. The adiabatic assumption allows one to construct a state-specific effective potential (or, channel potential) as a function of the separation between the fragments. This channel potential is the sum of the reaction coordinate potential (i.e., the minimum-energy pathway from reactant to products on the potential surface), the minimum rotational energy associated with the total angular momentum J , and the energy associated with the given rovibrational quantum number (all of these quantities varying as a function of the separation). Thus, instead of determining the minimum sum of states variationally as in VRRKM, one counts the number of open channels, i.e., the number of eigenstates whose channel maxima along the reaction coordinate do not exceed the total energy E . The SACM is generally expected to produce rate coefficients less than or equal to those of VRRKM, as a result of the upper-bound property of VRRKM which arises from the use of the transition-state assumption.

It should be noted that, in principle, both the SACM and VRRKM require for their implementation the same information: the eigenvalues of the eigenstates for the modes orthogonal to the reaction coordinate as a function of the separation between the fragments in the flexible transition state. A major difference between the two theories is the way in which one uses these eigenstates to calculate a rate coefficient, as described above. The SACM has been formulated and implemented quantum mechanically,^{3,21,22} whereas recent developments in VRRKM have tended to use a combination of quantum and classical state-counting methods.^{2,23} In the classical evaluation of a sum of states, one determines the available phase-space volume in units of Planck's constant h for a given maximum energy E and a fixed angular momentum J . This is evaluated by an n -dimensional phase-space integral (see, e.g., Ref. 16) over an appropriate set of coordinates $(q_1, \dots, q_n)$ and their conjugate momenta $(p_1, \dots, p_n)$ as follows:

$$W_{\text{class}}(E, J) = h^{-n} \int \cdots \int dq dp S[E - H] \delta[j - J], \quad (2)$$

where $dq = dq_1 \cdots dq_n$, $dp = dp_1 \cdots dp_n$, n is the number of degrees of freedom involved, H is the classical Hamiltonian for these degrees of freedom, and j is the total system angular momentum for a given set of p values. $S[E - H]$ is the Heaviside step function, $S[x] = 0, x \leq 0$ and $S[x] = 1, x \geq 0$, which imposes the constraint of a maximum total energy E , and $\delta[j - J]$ is the Dirac δ function which ensures that only momentum values p for which the total angular momentum j is equal to J can contribute to the integral.

The second problem is that of correctly accounting for angular momentum conservation in the rate coefficient calculation. Within the statistical framework, this requires one to evaluate the sum of states $W(E, J)$ of the transition state and the density of states $\rho(E, J)$ of the reactant molecule with due regard for the constraints imposed by a fixed total angular momentum J . Fast and accurate methods of evaluating the density of states of the reactant molecule $\rho(E, J)$ have previously been elaborated,^{20,2} and this aspect of the problem will not be considered in the present paper. The first step in imposing a fixed total angular momentum on the calculated sum of states, $W(E, J)$ is to identify which degrees of freedom in the system entail motion that contributes significantly to J . In a flexible transition state, the internal vibrations of the fragments are, in most cases, unlikely to make any significant contribution to J . Therefore, one need consider only the rotations of the fragments and the orbital motion when applying the fixed- J constraint. These rotational degrees of freedom are collectively called the *transitional modes* (e.g., Ref. 2), since they change qualitatively in nature during passage from the reactant molecule to the flexible transition state and then to the completely separated product fragments. Their sum of states, $W_{\text{TM}}(E, J)$ is written explicitly as a function of the angular momentum. The quantum vibrational density of states, $\rho_{\text{vib}}(E)$ is easily and rapidly evaluated via the Beyer–Swinehart algorithm,²⁴ and may be folded in with the transitional-mode sum of states $W_{\text{TM}}(E, J)$ by standard convolution methods (e.g., Ref. 16) in order to obtain the overall sum of states $W(E, J)$:

$$W(E, J) = \int_{E_{\text{min}}(J)}^{E^*} \rho_{\text{vib}}(E^* - E') W_{\text{TM}}(E', J) dE', \quad (3)$$

where E^* is the maximum energy available for distribution among the degrees of freedom orthogonal to the reaction coordinate:

$$E^* = E - V(r), \quad (4)$$

where r is the separation between the fragment centers of mass, $V(r)$ is the reaction coordinate potential (i.e., the minimum-energy pathway), and E is taken as referenced to the zero-point vibrational energy of the reactant molecule. In Eq. (3), $E_{\text{min}}(J)$ is the minimum energy required to generate the angular momentum J (this will be further discussed later). Alternatively, one may obtain $W(E, J)$ by convoluting the vibrational *sum* of states with the transitional-mode *density* of states.²

The simplest way to allow for the constraint imposed by a fixed angular momentum J on the transitional-mode sum of states $W_{\text{TM}}(E, J)$ in a flexible transition state is to make the approximation that $J \approx L$, where L is the orbital angular momentum, and therefore to conserve L and treat the fragment rotations as independent, internal modes.²⁵ This method, while of considerable practical utility, does not always suffice, especially when applied to reactions involving heavy fragments.²⁶ The more general angular momentum problem was first attacked in a rigorous fashion in the phase-space theory (PST) developed by Pechukas and Light²⁷ and Nikitin.²⁸ In PST, the exact energy- and angular-momentum-resolved rotational sum of states, $W_{\text{PST}}(E, J)$, is evaluated for fragments interacting with a central potential (i.e., freely

rotating fragments) by transformation to the appropriate rotational action-angle variables and either counting the quantum states directly or evaluating the classical phase-space volume integral. The transition state (often called the orbiting transition state, e.g., Ref. 29), is placed at the position of the centrifugal barrier induced by the conservation of the orbital angular momentum L . PST has been developed, in quantum and classical forms, to apply to all the relevant combinations of atomic, linear, spherical-top, and symmetric-top fragments.³⁰ PST is limited in its applicability to systems with a *central* potential of interaction because of the assumption of completely product-like, freely rotating fragments in the orbiting transition state. This assumption is, unfortunately, a poor one for most reactions, the notable exception being reactions between an ion and an isotropic-induced dipole. In the presence of a noncentral potential, there are barriers to rotation of the fragments. This causes a significant reduction in $W_{\text{TM}}(E, J)$ and therefore reduces the predicted rate coefficient relative to the PST result.^{31,32} The presence of noncentrality in the potential not only reduces the absolute rate coefficient, but also shifts the variationally selected transition states (or, the channel maxima) to separations of the fragments smaller than those of the centrifugal barriers.

The third problem, which is intimately related to the second, is precisely how to account for the loose, anharmonic hindrance to the fragment rotations in flexible transition states while still maintaining rigorous angular momentum conservation. The " $J \approx L$ " approximation offers a comparatively simple solution to this problem: since constraints imposed by angular momentum conservation on the fragment rotations are neglected, one requires simply sums of states $W(E)$ (or, channel eigenvalues) for the "internal" hindered fragment rotations. Analytic solutions for the sum of states or channel eigenvalues of hindered internal rotors with a number of model potentials are available.^{33,26,34,31,35,36} Numerical methods have also been applied for arbitrary hindering potentials at the more approximate canonical level,³⁷ where one requires only partition functions as opposed to the more-detailed microscopic quantities. However, as mentioned above and will be illustrated in the present paper, the $J \approx L$ approximation breaks down under certain circumstances, and a more complete method for dealing with the problem is desirable.

One rigorous way to approach the problem of evaluating $W_{\text{TM}}(E, J)$ for noncentral interaction potentials is via the use of action-angle variables.⁴ Augustine and Miller³⁸ pioneered the use of action-angle variables for bimolecular systems where the two rotating molecules interact with a noncentral potential. In this approach, the rotational action-angle variables of PST are used, with the noncentral interaction potential being expressed in terms of the angle variables. This is carried out via successive coordinate transformations: from those initial coordinates in which it is convenient to express the model potential, to the conventional Euler coordinates, and finally to the angle variables. Since one of the action variables is precisely the total angular momentum J , evaluation of the phase-space volume integral in terms of the action-angle variables subject to the energy constraint

and the angular momentum triangle inequalities (e.g., Ref. 2) automatically ensures that the total angular momentum is conserved. This approach was brought to its full fruition in the work of Wardlaw and Marcus,^{39,2,40} who evaluated the necessary phase-space integral numerically with a Monte Carlo technique. The principal disadvantage associated with such an approach is that the coordinate transformations from the first set of coordinates, in which it is convenient to express the potential, to the final angle variables are very complex. This means that the functional form of the potential used in the phase-space integral is also complex, such that the integral must be evaluated numerically over the full phase-space dimensionality (minus those momenta which can be eliminated due to conservation requirements and their conjugate cyclic coordinates²). Klippenstein and Marcus²³ have introduced a faster numerical integration technique by returning to conventional Euler coordinates and momenta, in which the potential takes a much simpler form, for the evaluation of the phase-space integral. Nevertheless, the computational time required is still considerable compared with VRRKM calculations using the more approximate $J \approx L$ approximation [the Klippenstein–Marcus algorithm requires 10–15 min on a VAX 11/780 for the calculation of a single $k(E, J)$,²³ whereas a VRRKM calculation using the $J \approx L$ approximation (e.g., Ref. 16) requires about the same time for the calculation of the entire thermal rate coefficient at 300 K]. Furthermore, when expressed in terms of Euler coordinates and momenta, the phase-space integral has not proved amenable to analytical simplification.

Troe^{20,31} has developed simplified methods of dealing with the constraint of total angular momentum conservation within the framework of the SACM. In this approach one starts with the exact eigenvalues of PST for a central potential and then proceeds to introduce "rigidity factors" which represent the effect on the predicted rate coefficient of the modifications induced by the noncentrality of the potential in the channel maxima of PST. The rigidity factors are obtained using channel eigenvalues based on perturbation treatments of model anharmonic and sinusoidal forms for the potential hindering the rotations of the fragments.^{34,31} The effects of total angular momentum conservation are incorporated by focusing on the limiting low- and high- J behavior of the channel eigenvalues.

In this work, a new method for the classical evaluation of $W_{\text{TM}}(E, J)$ is presented which substantially simplifies the task of calculating the microscopic rate coefficient. The transitional-mode phase-space integral is evaluated in two stages. First, one transforms the momentum-space integral in such a way that accurate analytic expressions for the available momentum-space volume $\Phi(E, J, \mathbf{q})$ for a given configuration $\mathbf{q}$ may be obtained. These expressions apply to all relevant combinations of atom, linear, spherical-top, symmetric-top, and asymmetric-top fragments. In the case of spherical-top fragments, the expression for $\Phi(E, J, \mathbf{q})$ is exact. Second, $W_{\text{TM}}(E, J)$ is obtained by integrating $\Phi(E, J, \mathbf{q})$ over the available configuration space. The integration over configuration space may be carried out with different levels of approximation. Exact account for the

bonded and nonbonded interactions requires at most a five-dimensional integral. In the present work, we derive accurate analytic approximations to $W_{\text{TM}}(E, J)$ by using a simple model potential for the fragment interactions which incorporates the principal features of more-accurate potentials: noncentrality with an appropriate barrier $V_0(r)$ to the fragment rotations, coupling between the configurations of the two fragments (the rotors are *not* assumed to rotate under independent potentials as in simpler treatments, e.g., Refs. 41, 42, 37, 26, and 31), and nonbonded steric interactions.

The analytic expressions obtained for $\Phi(E, J, \mathbf{q})$ have the additional advantage of highlighting the sources of error inherent in the simpler " $J \approx L$ " model. It can immediately be seen that the simpler model breaks down under two circumstances: (a) at low J , the model can substantially overestimate the available momentum-space volume due to the neglect of the restriction $-J \leq K \leq J$, where K is the projection of total angular momentum vector $\mathbf{J}$ onto the axis joining the two fragment centers of mass, and (b) the model can substantially overestimate $\Phi(E, J, \mathbf{q})$ for *all* values of J when applied to reactions involving heavy fragments since, in the expression for the rotational kinetic energy, it treats the fragment rotations as independent rather than using properly reduced rotational constants. Both of these problems have previously been identified;^{20,26} the present analysis provides a transparently clear and straightforwardly applied means of resolving them at very little extra computational expense.

Clearly, the optimum method for evaluating $W_{\text{TM}}(E, J)$ is by calculation of the exact quantum eigenstates as a function of the separation between the fragments (as has been done for some very simple systems, e.g., Ref. 43). However, it would seem that a classical evaluation of the sum of states via the appropriate phase-space integral is sufficiently accurate for most purposes, with the exception of very low temperatures (e.g., Refs. 34, 31, and 44). Evidence to date suggests that the necessary quantum correction to the classical sum of states for the transitional modes is small under most conditions.^{45,42}

The present work is structured as follows. In Sec. II, the general technique is outlined and applied in detail for the derivation of $\Phi(E, J, \mathbf{q})$ for the simplest case: that of a flexible transition state with two spherical-top fragments. The connection between the present approach and the action-angle variable formalism^{2,30,20} is established and discussed (see also Appendix A). In addition, the sources of error in the simple " $J \approx L$ " approximation are delineated and the necessary corrections identified. In Sec. III, the dependence of $\Phi(E, J, \mathbf{q})$ on inertial asymmetry of the fragments is examined in detail by the application of the method to a flexible transition state with two linear fragments. Concise, generalized expressions for $\Phi(E, J, \mathbf{q})$ which apply to all relevant combinations of atom, linear, spherical-top, symmetric-top, and asymmetric-top fragments are thereby derived. The integration of $\Phi(E, J, \mathbf{q})$ over the configuration space to generate $W_{\text{TM}}(E, J)$ is considered in Sec. IV. Assumptions which lead to analytic approximations to $W_{\text{TM}}(E, J)$ are discussed, and the results for some useful model potentials compared. The work is concluded in Sec. V.

II. GENERAL THEORY: APPLICATION TO THE SPHERICAL-TOP-SPHERICAL-TOP LOOSE TRANSITION STATE

In the present section, an exact expression for the available momentum-space volume $\Phi(E, J, \mathbf{q})$ for a fixed configuration $\mathbf{q}$ of a flexible transition state containing two spherical-top fragments at a center-of-mass separation r will be derived. This enables the basic method to be outlined in detail using the simple Hamiltonian for the spherical-top-spherical-top system [Eq. (6)]. The extension of the method to systems involving inertially asymmetric fragments is presented in the next section. The classical sum of states for the transitional modes of a flexible transition state consisting of two spherical-top fragments may be written in terms of the conventional Euler angles and their conjugate momenta as.²³

$$W_{\text{TM}}(E, J) = \frac{1}{h^{n_{\sigma_1 \sigma_2}}} \int \cdots \int d\theta_1 d\phi_1 d\psi_1 d\theta_2 d\phi_2 d\psi_2 \\ \times d\theta d\phi dP_{\theta_1} dP_{\phi_1} dP_{\psi_1} dP_{\theta_2} dP_{\phi_2} dP_{\psi_2} \\ \times dP_{\theta} dP_{\phi} S[E^* - H_{\text{TM}}] \delta[j - J], \quad (5)$$

where $(\theta_1, \phi_1, \psi_1, P_{\theta_1}, P_{\phi_1}, P_{\psi_1})$ are the Euler angles and their conjugate momenta for fragment 1, $(\theta_2, \phi_2, \psi_2, P_{\theta_2}, P_{\phi_2}, P_{\psi_2})$ are those for fragment 2, and $(\theta, \phi, P_{\theta}, P_{\phi})$ are those for the two-dimensional orbital rotor (i.e., the relative orbital motion with moment of inertia μr^2 , where μ is the reduced mass of the fragment pair). σ_1 and σ_2 are the rotational symmetry numbers of the two fragments, included to avoid the multiple counting of indistinguishable configurations. H_{TM} is the transitional-mode Hamiltonian [Eq. (6)], E^* is defined as in Eq. (4), and j is the total angular momentum associated with the given values of the momenta. The Heaviside step function $S[E - H]$ ensures that only classical states for which $H \leq E$ are counted in the integral ($S[x] = 0$ for $x < 0$, and $S[x] = 1$ for $x \geq 0$). The Dirac δ -function $\delta[j - J]$ ensures that only momentum states for which the total angular momentum j is equal to J contribute to the phase-space volume integral. The Euler angles for each of the fragment rotors and for the orbital motion are referenced to the same spatially fixed set of Cartesian axes. The transitional-mode Hamiltonian H_{TM} is written in terms of these Euler angles and their conjugate momenta as

$$H_{\text{TM}} = \frac{1}{2I_1} \left[P_{\theta_1}^2 + \frac{(P_{\phi_1} - P_{\psi_1} \cos \theta_1)^2}{\sin^2 \theta_1} + P_{\psi_1}^2 \right] \\ + \frac{1}{2\mu r^2} \left(P_{\theta}^2 + \frac{P_{\phi}^2}{\sin^2 \theta} \right) + \frac{1}{2I_2} \\ \times \left[P_{\theta_2}^2 + \frac{(P_{\phi_2} - P_{\psi_2} \cos \theta_2)^2}{\sin^2 \theta_2} + P_{\psi_2}^2 \right] + V(\mathbf{q}) \\ = T_{\text{rot}} + V(\mathbf{q}), \quad (6)$$

where I_i is the moment of fragment i , μr^2 is that for the orbital rotation, T_{rot} is the rotational kinetic energy, and $V(\mathbf{q})$ is the potential for the transitional-mode coordinates. As noted in the Introduction, the kinetic energy and angular momentum relationships are sufficiently complex when ex-

pressed in terms of the Euler angles and their conjugate momenta that one cannot easily make headway in seeking analytic simplifications to Eq. (5). These relationships are simplified by a transformation to the principal-axis momenta for each of the rotors concerned:

$$\{P_{\theta 1}, P_{\phi 1}, P_{\psi 1}, P_{\theta 2}, P_{\phi 2}, P_{\psi 2}, P_{\theta}, P_{\phi}, P_{\psi}\} \\ \xrightarrow{J_c = \sin \theta_1 \sin \theta_2 \sin \theta} \{j'_{1x} j'_{1y} j'_{1z} j'_{2x} j'_{2y} j'_{2z}, l_x, l_y\}, \quad (7)$$

where $(j'_{1x} j'_{1y} j'_{1z})$ are the principal-axis momenta of fragment i (i.e., the projections of the angular momentum vector $\mathbf{j}_i$ onto the principal axes of the fragment), and (l_x, l_y) are those of the orbital rotor (the fragment principal-axis momenta are written with a prime and those of the orbital-rotor without for later notational convenience). J_c is the Jacobian for the transformation, which is noncanonical since the original Euler coordinates are retained and only the momenta transformed; J_c is evaluated according to the transformation relations:

$$j'_{ix} = \frac{\sin \psi_i}{\sin \theta_i} (P_{\phi i} - P_{\psi i} \cos \theta_i) + P_{\theta i} \cos \psi_i, \quad (8a)$$

$$j'_{iy} = \frac{\sin \psi_i}{\sin \theta_i} (P_{\phi i} - P_{\psi i} \cos \theta_i) + P_{\theta i} \sin \psi_i, \quad (8b)$$

$$j'_{iz} = P_{\psi i}, \quad (8c)$$

for $i = 1, 2$, and

$$l_x = P_{\theta}, \quad (9a)$$

$$l_y = P_{\phi} / \sin \theta, \quad (9b)$$

where the Euler angles are defined such that the line of nodes is the intermediate x axis.⁴⁶ The rotational kinetic energy is written in terms of the new momenta as

$$T_{\text{rot}} = \frac{1}{2I_1} (j'^2_{1x} + j'^2_{1y} + j'^2_{1z}) + \frac{1}{2I_2} (j'^2_{2x} + j'^2_{2y} + j'^2_{2z}) \\ + \frac{1}{2\mu r^2} (l_x^2 + l_y^2), \quad (10)$$

and the $W_{\text{TM}}(E, J)$ is written as

$$W_{\text{TM}}(E, J) = \frac{1}{h^n \sigma_1 \sigma_2} \int \cdots \int d\theta_1 d\phi_1 d\psi_1 d\theta_2 d\phi_2 d\psi_2 d\theta d\phi \\ \times \sin \theta_1 \sin \theta_2 \sin \theta dj'_{1x} dj'_{1y} dj'_{1z} dj'_{2x} dj'_{2y} dj'_{2z} \\ \times dj'_{2z} dl_x dl_y S[E^* - H_{\text{TM}}] \delta[j - J]. \quad (11)$$

It is convenient to allow the angular momenta to take units of $h/2\pi$. Written in this way they are dimensionless quantities, and the kinetic energy takes the form

$$T_{\text{rot}} = B_1 (j'^2_{1x} + j'^2_{1y} + j'^2_{1z}) + B_2 (j'^2_{2x} + j'^2_{2y} + j'^2_{2z}) \\ + B_0 (l_x^2 + l_y^2), \quad (12)$$

where $B_1 = h^2/8\pi^2 I_1$, $i = 1, 2$, and $B_0 = h^2/8\pi^2 \mu r^2$. $W_{\text{TM}}(E, J)$ is then written as

$$W_{\text{TM}}(E, J) = \frac{1}{\sigma_1 \sigma_2} \frac{1}{(2\pi)^8} \int \cdots \int d\theta_1 d\phi_1 d\psi_1 d\theta_2 d\phi_2 d\psi_2 \\ \times d\theta d\phi \sin \theta_2 \sin \theta_1 \sin \theta dj'_{1x} dj'_{1y} dj'_{1z} \\ \times dj'_{2x} dj'_{2y} dj'_{2z} dl_x dl_y S[E^* - H_{\text{TM}}] \\ \times \delta[j - J]. \quad (13)$$

We now define the available momentum-space volume for a given configuration $\mathbf{q}$, $\Phi(E, J, \mathbf{q})$, as

$$\Phi(E, J, \mathbf{q}) = \frac{1}{\pi^3} \int \cdots \int dj'_{1x} dj'_{1y} dj'_{1z} dj'_{2x} dj'_{2y} dj'_{2z} \\ \times dl_x dl_y S[E^* - H_{\text{TM}}] \delta[j - J] \quad (14)$$

so that

$$W_{\text{TM}}(E, J) = \frac{1}{\sigma_1 \sigma_2} \frac{1}{N} \int \cdots \int d\theta_1 d\phi_1 d\psi_1 d\theta_2 d\phi_2 d\psi_2 d\theta d\phi \\ \times \sin \theta_1 \sin \theta_2 \sin \theta \Phi(E, J, \mathbf{q}) \\ = \langle \Phi(E, J, \mathbf{q}) \rangle, \quad (15)$$

where $N = 2^8 \pi^5$ is the normalizing factor for the configuration-space integral. As indicated in Eq. (15), $W_{\text{TM}}(E, J)$ may formally be regarded as the average of $\Phi(E, J, \mathbf{q})$ over all possible configurations of the system at the given center-of-mass separation r .

The total angular momentum vector $\mathbf{j}$ is related to the individual angular momenta by the vector addition:

$$\mathbf{j} = \mathbf{j}_1 + \mathbf{j}_2 + \mathbf{l}. \quad (16)$$

The constraints on the allowed combinations of the principal-axis momenta in Eq. (14) are therefore implicit in Eq. (16). Before Eq. (16) can be applied, however, it is necessary to resolve the individual angular momentum vectors onto a common set of Cartesian axes. It is convenient to use the principal axes of the orbital rotor for this purpose (these are related to the spatially fixed Cartesian axes though the orbital Euler coordinates θ and ϕ). The principal-axis components $(j'_{ix} j'_{iy} j'_{iz})$ of the fragment angular momentum $\mathbf{j}_i$ are therefore transformed to the corresponding components $(j_{ix} j_{iy} j_{iz})$ relative to the *orbital* principal axes,

$$\{j'_{1x} j'_{1y} j'_{1z} j'_{2x} j'_{2y} j'_{2z}\} \\ \xrightarrow{J_c = 1} \{j_{1x} j_{1y} j_{1z} j_{2x} j_{2y} j_{2z}\}. \quad (17)$$

The Jacobian of this transformation is unity, since a rotation of Cartesian axes is an orthogonal transformation (e.g., Ref. 46). As a result of the symmetry of the spherical-top fragments, the kinetic energy retains the same form as in Eq. (12):

$$T_{\text{rot}} = B_1 (j^2_{1x} + j^2_{1y} + j^2_{1z}) + B_2 (j^2_{2x} + j^2_{2y} + j^2_{2z}) \\ + B_0 (l_x^2 + l_y^2), \quad (18)$$

and the available momentum-space volume is

$$\Phi(E, J, \mathbf{q}) = \frac{1}{\pi^3} \int \cdots \int dj_{1x} dj_{1y} dj_{1z} dj_{2x} dj_{2y} dj_{2z} \\ \times dl_x dl_y S[E^* - H_{\text{TM}}] \delta[j - J]. \quad (19)$$

It is desirable to utilize Eq. (16) to transform the momenta to a new set which is comprised of the body-fixed components j_x, j_y, j_z of the *total* angular momentum $\mathbf{j}$, and a set of *internal* momenta which, by definition, do not contribute to the total angular momentum, and for which the kinetic-energy expression remains separable. This can be achieved by noting that Eqs. (16) and (18) are analogous to the translational kinetic-energy and linear momentum relationships for two- and three-body systems. Along the x and y

axes, the present case is equivalent to a colinear three-body system, whereas along the z axis it is equivalent to a two-body system (recall that, since the orbital rotation is two-dimensional, $l_z \equiv 0$ and so only the two fragments can contribute to the angular momentum along the z axis of the orbital rotor). The desired transformations can therefore be carried out using a combination of the “center-of-mass” and “skewed-coordinate” transformations (see, e.g., Ref. 16). In order to apply these transformations canonically, one uses the appropriate generator function (e.g., Ref. 46) to determine the coordinate and momentum transformation relations. In the present case, we utilize only the appropriate momentum transformation relations, retaining the original Euler angles for our coordinates.

The “center-of-mass” momentum transformation to internal momenta ($p_x, p'_x, p_y, p'_y, p_z$) and the body-fixed components (j_x, j_y, j_z) of the total angular momentum is summarized as follows:

$$\{j_{1x}, j_{1y}, j_{1z}, j_{2x}, j_{2y}, j_{2z}, l_x, l_y\} \xrightarrow{J_c=1} \{j_x, j_y, j_z, p_x, p'_x, p_y, p'_y, p_z\}, \quad (20)$$

where j_x , j_y , and j_z are defined as in Eq. (16), and the internal momenta are defined by the transformation relations:

$$p_x = B_{012} \left[\frac{1}{B_1} (l_x + j_{2x}) - \frac{j_{1x}}{B_{02}} \right], \quad (21a)$$

$$p'_x = B_{012} \left[\frac{-1}{B_2} (l_x + j_{1x}) + \frac{j_{2x}}{B_{01}} \right], \quad (21b)$$

$$p_y = B_{012} \left[\frac{1}{B_1} (l_y + j_{2y}) - \frac{j_{1y}}{B_{02}} \right], \quad (21c)$$

$$p'_y = B_{012} \left[\frac{-1}{B_2} (l_y + j_{1y}) + \frac{j_{2y}}{B_{01}} \right], \quad (21d)$$

$$p_z = B_{12} \left(\frac{j_{1z}}{B_2} - \frac{j_{2z}}{B_1} \right), \quad (21e)$$

where $B_{012}^{-1} = B_0^{-1} + B_1^{-1} + B_2^{-1}$, and $B_{ij}^{-1} = B_i^{-1} + B_j^{-1}$. The kinetic energy becomes

$$T_{\text{rot}} = B_{012} (j_x^2 + j_y^2) + B_{12} j_z^2 + (B_1 + B_2) p_z^2 + (B_0 + B_1) (p_x^2 + p_y^2) + (B_0 + B_2) (p_x'^2 + p_y'^2) - 2B_{0x} p'_x - 2B_{0y} p'_y. \quad (22)$$

The kinetic energy associated with the internal momenta may be separated with the skewed-axis momentum transformation:

$$\{p'_x, p'_y\} \xrightarrow{J_c=\alpha^2} \{p''_x, p''_y\}, \quad (23)$$

where

$$p''_x = \frac{1}{\alpha} \left(p'_x - \frac{B_{02}}{B_2} p_x \right), \quad (24a)$$

$$p''_y = \frac{1}{\alpha} \left(p'_y - \frac{B_{02}}{B_2} p_y \right), \quad (24b)$$

and

$$\alpha^2 = \frac{B_1 B_{02}^2}{B_0 B_2 B_{012}}. \quad (25)$$

The kinetic energy finally becomes

$$T_{\text{rot}} = B_{012} (j_x^2 + j_y^2) + B_{12} j_z^2 + (B_1 + B_2) p_z^2 + \frac{B_1 B_{02}}{B_{012}} (p_x^2 + p_x'^2 + p_y^2 + p_y'^2). \quad (26)$$

We may now identify B_{012} and B_{12} , respectively, as the rotational constants B_{ext} and A_{ext} of the overall body (B_{ext} being degenerate and A_{ext} unique, since the overall body is a symmetric top):

$$B_{\text{ext}} = B_{012} = (B_0^{-1} + B_1^{-1} + B_2^{-1})^{-1}, \quad (27a)$$

$$A_{\text{ext}} = B_{12} = (B_1^{-1} + B_2^{-1})^{-1}. \quad (27b)$$

Equation (27b) is a commonly used result. The result of Eq. (27a) arises out of the present treatment, and may be proven directly in a straightforward manner. A proof of Eq. (27a) is provided in Appendix A. In Eq. (26), the quantity $(B_1 + B_2)$ is the rotational constant corresponding to the reduced moment of inertia commonly used for internal torsional rotation about the body-fixed z axis (e.g., Ref. 16). It is instructive to compare the exact kinetic-energy expression of Eq. (26) with the kinetic-energy expression T'_{rot} which corresponds to the simple “ $J \approx L$ ” model:

$$T'_{\text{rot}} = B_{\text{ext}} (j_x^2 + j_y^2) + B_1 (j_{1x}^2 + j_{1y}^2 + j_{1z}^2) + B_2 (j_{2x}^2 + j_{2y}^2 + j_{2z}^2) = B_{\text{ext}} (j_x^2 + j_y^2) + A_{\text{ext}} j_z^2 + B_1 (j_{1x}^2 + j_{1y}^2) + B_2 (j_{2x}^2 + j_{2y}^2) + (B_1 + B_2) p_z^2. \quad (28)$$

In Eq. (28) the individual fragment momenta j_{ix} , j_{iz} have been retained since, in the $J \approx L$ model the fragment rotations are treated as independent. As indicated, one can equivalently use the correct rotational constants A_{ext} and $(B_1 + B_2)$ for external and internal (torsioned) rotation about the body-fixed z axis. However, within the $J \approx L$ model this makes no difference to the eventual sum of states because the proper limits on j_z at low- J values ($-J \leq j_z \leq J$) are in any case neglected. Comparison of Eqs. (26) and (28) shows that the simple $J \approx L$ model is inconsistent in its treatment of the x - and y -axis rotations (i.e., the two-dimensional “rocking modes”). The correct external rotational constant B_{ext} is used, but the internal rotational constants correspond to moments of inertia which are not properly reduced (“reduced” is used here in the more general context of the three-body momentum transformations outlined above). This inconsistency can lead to serious error but, as will become apparent from the results of this section, is very easily corrected.

With the external and internal momenta of Eq. (26), the momentum-space volume $\Phi(E, J, \mathbf{q})$ is cast into a tractable form:

$$\Phi(E, J, \mathbf{q}) = \frac{\alpha^2}{\pi^3} \int \int \int dj_x dj_y dj_z \delta[j - J] \times \int \cdots \int dp_x dp_y dp'_x dp'_y dp_z S[E^* - H_{\text{TM}}] \quad (29)$$

where the Jacobian α^2 is as defined in Eq. (25). Note that the $\delta[j - J]$ function has now been removed outside the inte-

grals over the internal momenta since these momenta, by definition, do not contribute to the system's total angular momentum j . The integrals over the internal momenta in Eq. (29) are easily cast in the form of Dirichlet's integral (e.g., Ref. 47), and may be evaluated to give

$$\begin{aligned}\Phi(E, J, \mathbf{q}) &= \frac{8}{15\pi} \frac{B_{\text{ext}} (A_{\text{ext}})^{1/2}}{B_0} \frac{1}{B_1^{3/2} B_2^{3/2}} \\ &\times \int \int \int dj_x dj_y dj_z \delta[j - J] \\ &\times [E^* - V(\mathbf{q}) - B_{\text{ext}}(j_x^2 + j_y^2) - A_{\text{ext}} j_z^2]^{5/2},\end{aligned}\quad (30)$$

where Eqs. (27a) and (27b) have been utilized as appropriate. The remaining integrals may be conveniently evaluated by transforming to the total angular momentum j , its body-fixed z -axis projection $K (= j_z)$ and a projection angle ξ :

$$\{j_x j_y j_z\} \xrightarrow{J_c=j} \{j, K, \xi\}, \quad (31)$$

where

$$j^2 = j_x^2 + j_y^2 + j_z^2, \quad (32a)$$

$$K = j_z, \quad (32b)$$

$$\xi = \tan^{-1}(j_y/j_x). \quad (32c)$$

$\Phi(E, J, \mathbf{q})$ may then be written as

$$\begin{aligned}\Phi(E, J, \mathbf{q}) &= \frac{8}{15\pi} \frac{B_{\text{ext}} (A_{\text{ext}})^{1/2}}{B_0} \frac{1}{B_1^{3/2} B_2^{3/2}} \\ &\times \int_0^{2\pi} d\xi \int_0^\infty dj j \delta[j - J] \\ &\times \int_{-K_{\text{max}}}^{K_{\text{max}}} dK [E^* - V(\mathbf{q}) - B_{\text{ext}} j^2 \\ &- (A_{\text{ext}} - B_{\text{ext}}) K^2]^{5/2}.\end{aligned}\quad (33)$$

Evaluation of Eq. (33) leads to

$$\begin{aligned}\frac{\Phi(E, J, \mathbf{q})}{(2J+1)} &= \frac{4}{15} \frac{B_{\text{ext}} (A_{\text{ext}})^{1/2}}{B_0} \frac{1}{B_1^{3/2} B_2^{3/2}} \left\{ \frac{(2J+1)}{3} [E^* - V(\mathbf{q}) - A_{\text{ext}} J(J+1)]^{5/2} \right. \\ &+ \frac{5}{12} (2J+1) [E^* - V(\mathbf{q}) - B_{\text{ext}} J(J+1)] [E^* - V(\mathbf{q}) - A_{\text{ext}} J(J+1)]^{3/2} \\ &+ \frac{5}{8} (2J+1) [E^* - V(\mathbf{q}) - B_{\text{ext}} J(J+1)]^2 [E^* - V(\mathbf{q}) - A_{\text{ext}} J(J+1)]^{1/2} \\ &\left. + \frac{5}{4(A_{\text{ext}} - B_{\text{ext}})^{1/2}} [E^* - V(\mathbf{q}) - B_{\text{ext}} J(J+1)]^3 \sin^{-1} \left[\frac{(2J+1)(A_{\text{ext}} - B_{\text{ext}})^{1/2}}{2[E^* - V(\mathbf{q}) - B_{\text{ext}} J(J+1)]^{1/2}} \right] \right\}, \\ &A_{\text{ext}} J(J+1) \leq E^* - V(\mathbf{q}),\end{aligned}\quad (34a)$$

$$\begin{aligned}\frac{\Phi(E, J, \mathbf{q})}{(2J+1)} &= \frac{\pi}{6} \frac{B_{\text{ext}} (A_{\text{ext}})^{1/2}}{B_0 (A_{\text{ext}} - B_{\text{ext}})^{1/2}} \frac{1}{B_1^{3/2} B_2^{3/2}} [E^* - V(\mathbf{q}) - B_{\text{ext}} J(J+1)]^3, \\ &B_{\text{ext}} J(J+1) \leq E^* - V(\mathbf{q}) \leq A_{\text{ext}} J(J+1),\end{aligned}\quad (34b)$$

where the resulting formulas have been "requantized" by setting $2J \rightarrow (2J+1)$ and $J^2 \rightarrow J(J+1)$ in order to ensure that the solution is well behaved at $J=0$ (e.g., Ref. 20).

Equations (34a) and (34b) represent the exact classical momentum-space volume for a fixed configuration $\mathbf{q}$ of a flexible transition state comprising of two spherical-top fragments. Note that, since the fragments in this case are entirely symmetric, the *only* configurational dependence which enters into $\Phi(E, J, \mathbf{q})$ is through the interaction potential $V(\mathbf{q})$, which alters the available kinetic energy.

Since the factor $(2J+1)$ for the degeneracy of the projection of J onto an arbitrary spatially fixed polar axis always cancels out in the numerator and denominator of the expression for $k(E, J)$, it is useful to plot the function $\Phi(E, J, \mathbf{q})/(2J+1)$ rather than $\Phi(E, J, \mathbf{q})$ itself, since the former is representative of the actual quantity which determines the microscopic rate coefficient.

In Figs. 1(a)–1(c), $\Phi(E, J, \mathbf{q})/(2J+1)$ is plotted (curve I) as a function of the angular momentum J for three representative systems. Figure 1(a) represents a flexible transition state with two light spherical-top fragments. The fragment rotational constants are taken to be the spherical average of those for a methyl radical, i.e., $B_i = (B_{\text{CH}_3^2} A_{\text{CH}_3})^{1/3}$. The reduced mass μ is that appropriate for two methyl fragments, and the separation r between the two centers of mass is given a value of 4.5 Å, which is typical of a canonical variational transition state for such a reaction at 300 K (e.g., Ref. 26). Figure 1(b) is representative of a heavy-heavy fragment combination, the spherically averaged rotational constants B_i and the reduced mass μ being appropriate to the CCl_3 radical. The separation is again 4.5 Å. Figure 1(c) is representative of a light-heavy fragment combination: B_1 is the spherically averaged rotational constant for CH_3 and B_2 is that for CCl_3 , with μ being

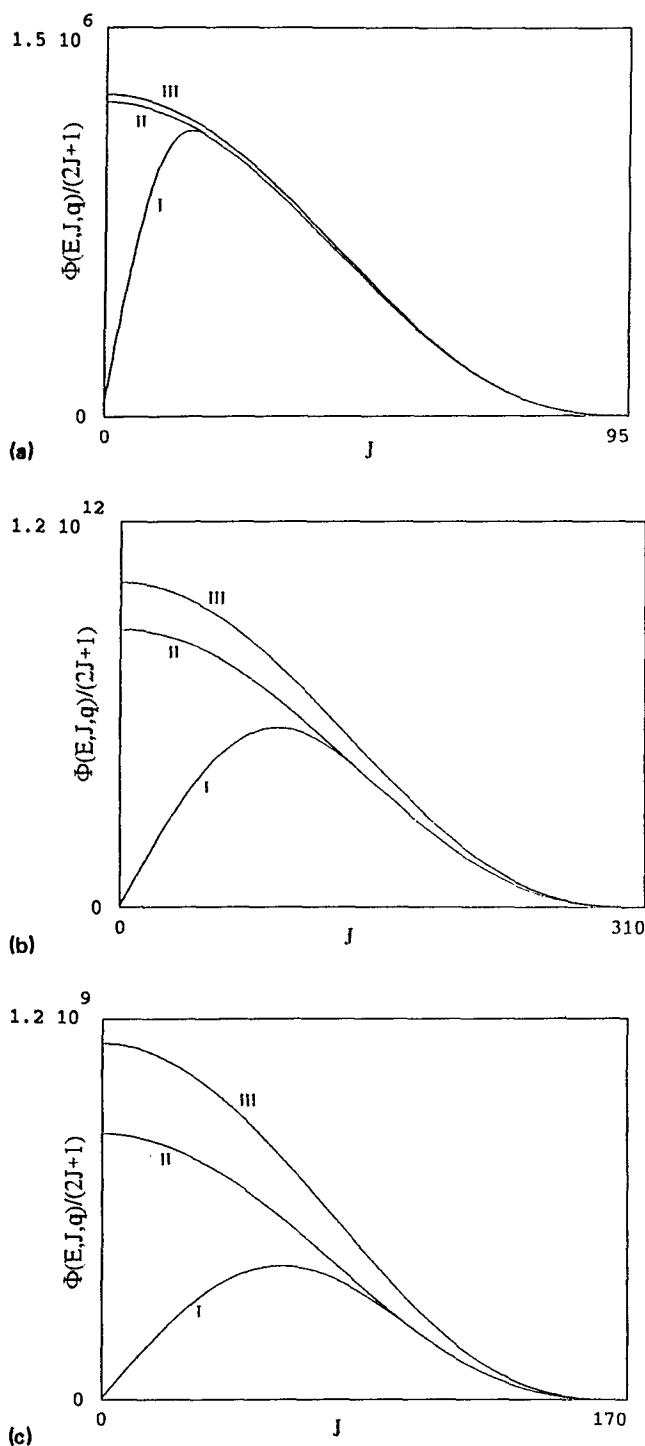


FIG. 1. Plot of momentum-space volume $\Phi(E, J, q)/(2J+1)$ [Eq. (34)] vs angular momentum J for (a) $\text{CH}_3 + \text{CH}_3$, (b) $\text{CCl}_3 + \text{CCl}_3$, and (c) $\text{CH}_3 + \text{CCl}_3$ systems. Center-of-mass separation 4.5 Å. Rotational constants spherically averaged from $A_{\text{CH}_3} = 4.73 \text{ cm}^{-1}$ and $B_{\text{CH}_3} = 9.45 \text{ cm}^{-1}$. Available kinetic energy $E^* - V(q) = 1000 \text{ cm}^{-1}$. Curve I: Exact low- J result of Eq. (34a). Curve II: Exact high- J result of Eq. (34b). Curve III: " $J \approx L$ " results [Eq. (35)].

evaluated accordingly. The separation r is 4.5 Å. The available kinetic energy, $E^* - V(q) = E - V(r) - V(q)$, is set at 1000 cm^{-1} ($2.86 \text{ kcal mol}^{-1}$) in each case [the shapes of the $\Phi(E, J, q)$ curves are qualitatively the same for different energies].

One can initially observe that the shapes of the curves in Fig. 1 are qualitatively the same as those obtained from PST.³⁰ The quantitative relationship between the present result and that of PST for the sphere-sphere system is considered below and in Appendix B.

Also plotted in Fig. 1 are two additional approximations of $\Phi(E, J, q)$. Curve II is the result obtained by neglect of the low- J limits on K , viz. $-J \leq K \leq J$. In this approximation, the "high- J " form of $\Phi(E, J, q)$, Eq. (34b), would be used for all values of J . Curve III is the result obtained from the simple $J \approx L$ approximation, which treats the fragment rotations as independent internal rotors. Beginning with the kinetic-energy expression for this model, Eq. (28), and pursuing the same integrations as above, the $J \approx L$ result for $\Phi(E, J, q)$ may be written as

$$\frac{\Phi(E, J \approx L, q)}{(2J+1)} = \frac{\pi}{6} \frac{1}{B_1^{3/2} B_2^{3/2}} [E^* - V(q) - B_{\text{ext}} J(J+1)]^3. \quad (35)$$

From Fig. 1, it is clear that the $J \approx L$ model (curve III) has two sources of error. The first source of error produces the overestimate of curve III relative to curve II, and is related to the treatment of the x - and y -axis internal rotations (the "rocking" modes). As noted earlier, the rotational constants used involve moments of inertia which are not properly reduced. The magnitude of the error is indicated by the ratio of Eq. (35) to Eq. (34b):

$$\frac{\Phi(E, J \approx L, q)}{\Phi(E, J, q)} = \frac{B_0(A_{\text{ext}} - B_{\text{ext}})^{1/2}}{B_{\text{ext}}(A_{\text{ext}})^{1/2}}. \quad (36)$$

Hence, the simple $J \approx L$ model *always* overestimates the momentum space. This overestimate is most serious where heavy fragments are involved, e.g., in the case of $\text{CH}_3 + \text{CCl}_3$ at $r = 4.5 \text{ Å}$ the sum of states at high J is overestimated by ca. 34%, and in the case of $\text{CCl}_3 + \text{CCl}_3$ the overestimate is ca. 16%.

The second source of error is the neglect at low- J values of the limitation on the allowed values of K , viz. $-J \leq K \leq J$. As is apparent from Fig. 1, this leads to a substantial overestimate of the available momentum space at low- J values. Although this error appears dramatic, its importance is tempered in the evaluation of high-pressure-limiting unimolecular or recombination rate coefficients by the fact that the reactant Boltzmann population introduces an additional degeneracy factor of $(2J+1)$, which reduces the importance of the low- J values. This is illustrated in Fig. 2, where $\Phi(E, J, q)$ for the system of Fig. 1(b) is plotted. In addition, the neglect of this same low- J constraint in the evaluation of the molecular density of states $\rho(E, J)$, which appears in the denominator of the expression for $k(E, J)$, will tend to cancel (or, indeed, overcompensate for) the corresponding error in the sum of states. Nevertheless, for comparison with low- J , state-resolved experimental measurements and for calculating thermal rate coefficients in the falloff regime, where non-equilibrium populations increase the importance of low- J values, it is necessary to allow for the proper constraints of K .

For an exact evaluation of the sum of states, one should properly use Eq. (34a) for the low- J momentum-space vol-

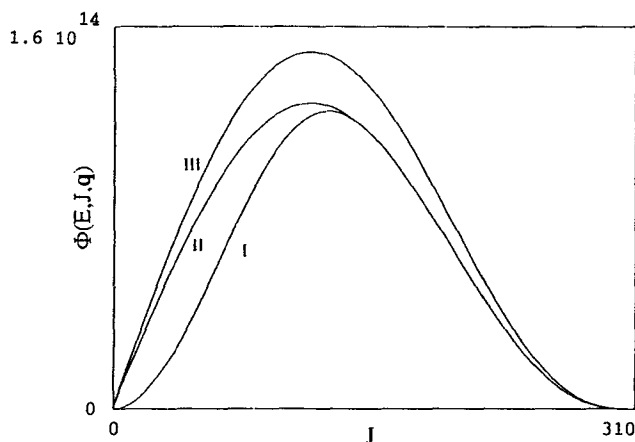


FIG. 2. Plot of $\Phi(E, J, q)$ vs angular momentum J for the $\text{CCl}_3 + \text{CCl}_3$ system of Fig. 1(b).

ume, followed by an exact numerical evaluation of $W_{\text{TM}}(E, J)$ [Eq. (15)] with inclusion of an appropriate potential $V(q)$ for the transitional modes. It should be noted that, since three angles may always be associated with motion of the overall body and hence are cyclic (i.e., do not appear in the potential), the maximum dimensionality for the numerical integral is five, so that one still obtains a substantial saving in computational time as compared with earlier methods.^{2,23} In the present paper, however, we seek accurate analytic approximations to $W_{\text{TM}}(E, J)$ with the aim of dramatically simplifying the implementation of VRRKM. For this purpose, it is necessary to use a simpler analytic form to approximate $\Phi(E, J, q)$ at low- J values so that the ensuing configuration-space integration is analytically tractable.

In order to develop a suitable approximation to the exact result of Eq. (34), it is useful to examine its low- J limit:

$$\lim_{J \rightarrow 0} \frac{\Phi(E, J, q)}{(2J+1)} = (2J+1) \frac{8}{15} \frac{B_{\text{ext}}(A_{\text{ext}})^{1/2}}{B_0} \times \frac{1}{B_1^{3/2} B_2^{3/2}} [E^* - V(q)]^{5/2}. \quad (37)$$

Figures 3(a)–3(c) illustrate two low- J approximations to $\Phi(E, J, q)$ for the three systems of Fig. 1. Curves I and II are the same as for Fig. 1. Curve III corresponds to the second of two low- J approximations suggested by Troe.²⁰ In this approximation, one simply extrapolates the low- J limit, Eq. (37) up to higher J values until it crosses the high- J result of curve II. This approximation works very well for light fragments [Fig. 3(a)], but not so well when heavy fragments are involved [Figs. 3(b) and 3(c)]. Curve IV represents an improved low- J approximation which takes some account of the effective potential induced by nonzero angular momenta with a spherically averaged rotational constant:

$$\frac{\Phi(E, \text{low } J, q)}{(2J+1)} = (2J+1) \frac{8}{15} \frac{B_{\text{ext}}(A_{\text{ext}})^{1/2}}{B_0} \frac{1}{B_1^{3/2} B_2^{3/2}} \times [E^* - V(q) - B_{\text{av}} J(J+1)]^{5/2}, \quad (38)$$

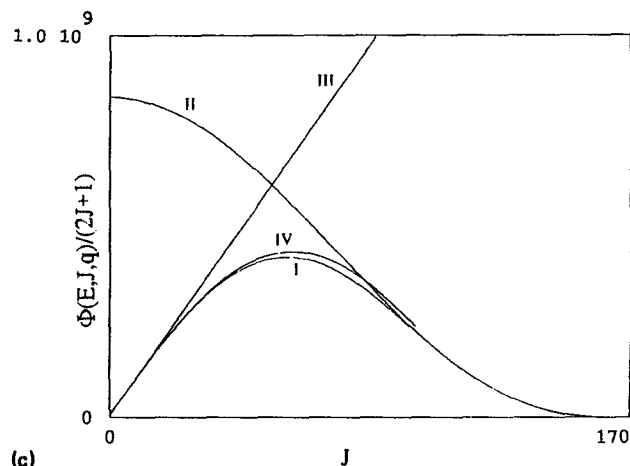
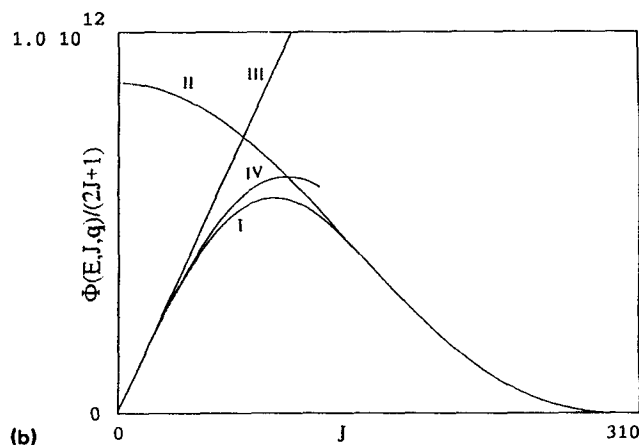
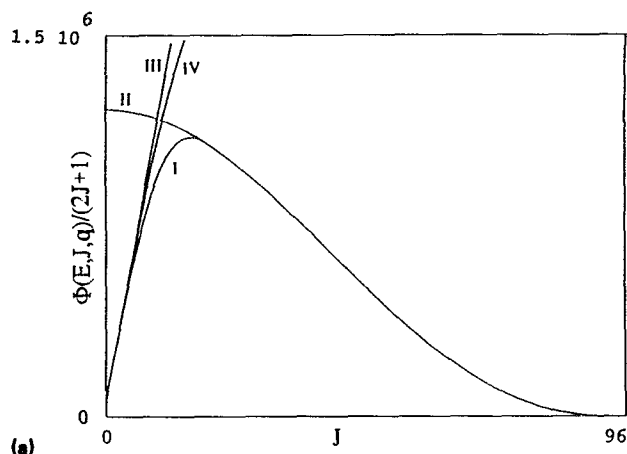


FIG. 3. Comparison of low- J approximations to $\Phi(E, J, q)$ with the exact result of Eq. (34a) for (a) $\text{CH}_3 + \text{CH}_3$ [parameters as in Fig. 1(a)], (b) $\text{CCl}_3 + \text{CCl}_3$ [parameters as for Fig. 1(b)], and (c) $\text{CH}_3 + \text{CCl}_3$ [parameters as for Fig. 1(c)]. Curve I: Exact result of Eq. (34a). Curve II: high- J limit, Eq. (34b). Curve III: Troe's low- J approximation, Eq. (37). Curve IV: Modified low- J approximation of Eq. (38).

where $B_{\text{av}} = (B_{\text{ext}}^2 A_{\text{ext}})^{1/3}$. As can be seen in Figs. 3(b) and 3(c), this approximation performs significantly better when heavy fragments are involved. An appropriate approximation to the exact results of Eqs. (34a) and (34b) which enables accurate analytic expressions for $W_{\text{TM}}(E, J)$ to be pursued (Sec. IV), is therefore

$$\Phi(E, J, q) = \min\{\Phi(E, \text{low } J, q), \Phi(E, \text{high } J, q)\}, \quad (39)$$

where $\Phi(E, \text{low } J, q)$ is prescribed in Eq. (38) and $\Phi(E, \text{high } J, q)$ is given by Eq. (34b).

Finally in this section, the consistency of the present approach, which applies noncanonical momentum transformations, with the canonical action-angle variable approach^{30,4} is established. This is most conveniently done by a comparison with phase-space theory in the limit of a central potential, for which we set $V(q) = 0$. Substituting Eqs. (34a) and (34b) into Eq. (15), the angular integrations cancel against the normalizing factor to yield the transitional-mode sum of states, $W_{\text{CP}}(E, J)$, of a sphere-sphere flexible transition state at a fragment separation r with a central potential of interaction:

$$W_{\text{CP}}(E, J) = \Phi(E, J) / \sigma_1 \sigma_2, \quad (40)$$

where $\Phi(E, J)$ is given by Eqs. (34a) and (34b) wherein $V(q)$ is set to zero. In principle, Eq. (40) could be applied variationally so that the minimum sum of states is determined. The result, however, would be larger than that of PST. The reason for this is that PST applies the additional dynamical constraint, appropriate to a central potential, that the orbital angular momentum L is a conserved quantity from some minimum fragment separation (assumed smaller than the range of separations where the transition states for the statistically important values of E and J lie) out to infinitely separated products. The presence of the L -dependent effective potential $L^2/2\mu r^2$ then reduces the sum of states to a value $W_{\text{PST}}(E, J) \leq W_{\text{CP}}(E, J)$.²⁷⁻³⁰

In the presence of a *noncentral* interaction potential, the orbital angular momentum L is no longer a conserved quantity (e.g., Ref. 48). Therefore, the additional energetic constraint imposed by an effective potential $L^2/2\mu r^2$ is dropped (e.g., Ref. 4). In order to establish consistency between the present approach and the canonical action-angle variable method, it is therefore necessary to modify the classical PST phase-space integral: one drops the L -conserving energetic constraint, which locates the orbiting transition state at the position r^* of the L -dependent centrifugal barrier, and imposes instead a fixed value of r for all values of L (Fig. 4). In Appendix B it is shown that, when the above modification to the PST phase-space integral is made, the action-angle variable approach also leads to the result of Eq. (40). This establishes the consistency of the two approaches.

III. EFFECTS OF INERTIAL ASYMMETRY IN THE FRAGMENT ROTORS: LINEAR, SYMMETRIC-TOP, AND ASYMMETRIC-TOP FRAGMENTS

The primary result of the preceding section, that of Eq. (34) for the total available momentum-space volume at a fixed configuration of a spherical-top-spherical-top flexible transition state, is classically exact. In the present section, the generalization of the method developed in Sec. II to cases involving inertially asymmetric rotors is considered. The momentum-space integrals for these more complex systems are generally complicated by angular coupling factors in the expression for the kinetic energy, and cannot be evaluated analytically for arbitrary configurations. However, accurate analytic approximations to $\Phi(E, J, q)$ can be obtained for a

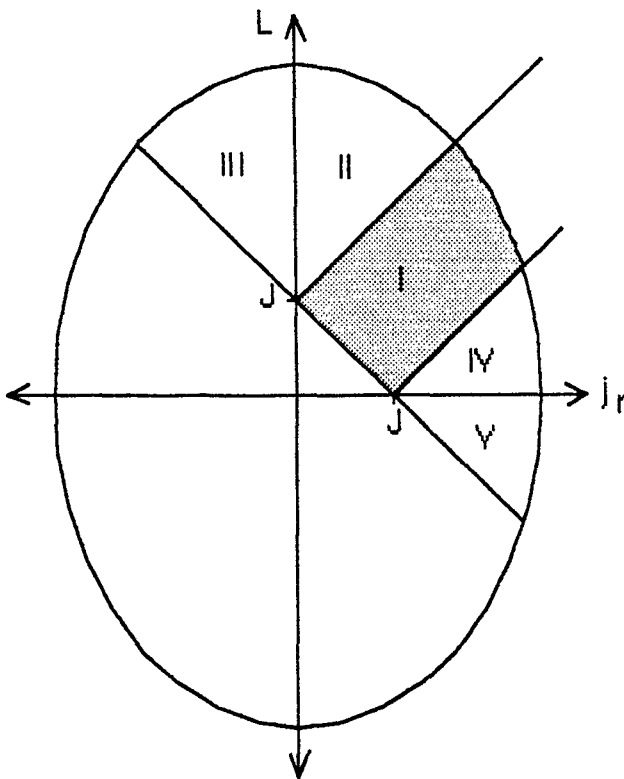


FIG. 4. Schematic plot of integration region in the (L, j_r) plane for the derivation of Eq. (34) by the method of action-angle variables (see Appendix B).

number of limiting configurations of the fragments. These known analytical formulas for limiting configurations admit straightforward interpolation so that arbitrary configurations can be very easily and accurately handled. It will be seen that the generalized expressions, which are no more difficult to evaluate than those of the preceding section, provide considerable insight into the dependence of the available momentum-space volume $\Phi(E, J, q)$ on changes in the inertial symmetry of the overall body with changing internal configuration of the fragments. For instance, it will be seen that in a system with two linear fragments, the function $\Phi(E, J, q)/(2J + 1)$ is dramatically different for a coaligned configuration as compared with an orthogonal configuration (Fig. 5). This introduces, for low- J values, a weighting in the sum of states which favors coalignment of the linear fragments, *in addition* to the effect of the noncentral potential $V(q)$ in favoring certain fragment orientations.

The procedure for extension of the method to linear and nonlinear asymmetric fragment rotors is straightforward. One observes that it is possible to evaluate $\Phi(E, J, q)$ analytically for a number of limiting configurations q_i . Examination of the function $\Phi(E, J, q_i)$ obtained in each case reveals that the general functional form is the same in all cases once the appropriate external rotational constants B_{ext} and A_{ext} of the overall body (approximated as a prolate top) are identified analogous to Eqs. (27a) and (27b) and Appendix A. A convenient solution for $\Phi(E, J, q)$ for *arbitrary* fragment orientations therefore presents itself: one simply allows B_{ext}

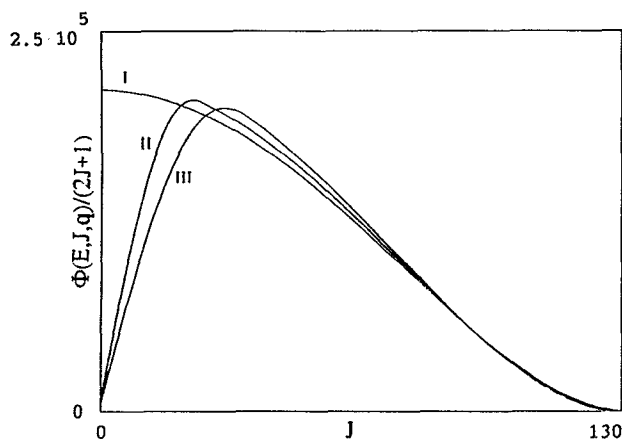


FIG. 5. Plot of $\Phi(E, J, \mathbf{q})$ for the HCN + HCN system at center-of-mass separation $4.5 \text{ \AA}$ and available kinetic energy $E^* - V(\mathbf{q}) = 1000 \text{ cm}^{-1}$. $B_{\text{HCN}} = 1.48 \text{ cm}^{-1}$. Curve I: collinear configuration. Curve II: fragment 1 aligned along line joining centers of mass and fragment 2 perpendicular to this line. Curve III: fragment 1, fragment 2, and line joining centers of mass all mutually perpendicular.

and A_{ext} to take values within the general functional form which are appropriate to any given configuration $\mathbf{q}$. This ensures that the correct functions $\Phi(E, J, \mathbf{q}_i)$ are obtained for the limiting configurations, and that there is a smooth interpolation of the function between these limiting forms.

The method will be illustrated for the case of a flexible transition state involving two linear fragments, since this system exhibits the maximum alteration in the inertial symmetry of the overall body with changing orientation of the fragments.

The classical sum of states $W_{\text{TM}}(E, J)$ for the transitional modes of a flexible transition state comprising of two linear fragments may be written analogous to Eq. (13) as

$$W_{\text{TM}}(E, J) = \frac{1}{\sigma_1 \sigma_2} \frac{1}{(2\pi)^6} \int \cdots \int d\theta_1 d\phi_1 d\theta_2 d\phi_2 d\theta d\phi \\ \times \sin \theta_1 \sin \theta_2 \sin \theta dj'_{1x} dj'_{1y} dj'_{2x} dj'_{2y} \\ \times dl_x dl_y S[E^* - H] \delta[j - J], \quad (41)$$

where in the present case the fragment principal-axis momenta j'_{1x} and j'_{2z} are zero due to the linearity of the rotors. The momentum-space volume integral $\Phi(E, J, \mathbf{q})$ is correspondingly written as

$$\Phi(E, J, \mathbf{q}) = \frac{1}{\pi^3} \int \cdots \int dj'_{1x} dj'_{1y} dj'_{2x} dj'_{2y} dl_x dl_y \\ \times S[E^* - H] \delta[j - J]. \quad (42)$$

As in the preceding section, it is necessary to resolve the individual angular momentum vectors $\mathbf{j}_1$, $\mathbf{j}_2$, and $\mathbf{l}$ onto a common set of Cartesian axes. This procedure is complex for arbitrary orientations of the linear fragments, since configurational coupling between the eventual internal and external angular momenta is introduced. However, such coupling does not appear for configurations wherein each linear fragment is aligned in the direction of one or other of the orbital principal axes. For the linear + linear system, there are five such distinct configurations, with corresponding kinetic-energy expressions as follows:

$$T_1 = B_1(j_{1x}^2 + j_{1y}^2) + B_2(j_{2x}^2 + j_{2y}^2) + B_0(l_x^2 + l_y^2), \quad (43a)$$

$$T_2 = B_1(j_{1x}^2 + j_{1y}^2) + B_2(j_{2y}^2 + j_{2z}^2) + B_0(l_x^2 + l_y^2), \quad (43b)$$

$$T_3 = B_1(j_{1y}^2 + j_{1z}^2) + B_2(j_{2x}^2 + j_{2y}^2) + B_0(l_x^2 + l_y^2), \quad (43c)$$

$$T_4 = B_1(j_{1y}^2 + j_{1z}^2) + B_2(j_{2y}^2 + j_{2z}^2) + B_0(l_x^2 + l_y^2), \quad (43d)$$

$$T_5 = B_1(j_{1y}^2 + j_{1z}^2) + B_2(j_{2x}^2 + j_{2z}^2) + B_0(l_x^2 + l_y^2). \quad (43e)$$

Analogous to the procedure of Sec. II, one carries out the necessary two- or three-body momentum transformations in each case to obtain external and internal angular momenta for which the kinetic-energy expression is separable. The integrals over the internal momenta and finally over j_z , ξ , and j may then be evaluated for each of the above cases to yield the following results.

For high J (in the collinear case, for all J):

$$\frac{\Phi(E, J, \mathbf{q}_i)}{(2J+1)} = \frac{B_{\text{ext}(i)} (A_{\text{ext}(i)})^{1/2}}{2B_0 (A_{\text{ext}(i)} - B_{\text{ext}(i)})^{1/2}} \frac{1}{B_1 B_2} \\ \times [E^* - V(\mathbf{q}) - B_{\text{ext}(i)} J(J+1)]^2, \\ B_{\text{ext}(i)} J(J+1) \leq E^* - V(\mathbf{q}) \leq A_{\text{ext}(i)} J(J+1), \quad (44a)$$

and, excepting the collinear case, for low J :

$$\frac{\Phi(E, J, \mathbf{q})}{(2J+1)} = \frac{B_{\text{ext}(i)} (A_{\text{ext}(i)})^{1/2}}{3\pi B_0} \frac{1}{B_1 B_2} \left((2J+1) [E^* - V(\mathbf{q}) - A_{\text{ext}(i)} J(J+1)]^{3/2} + \frac{3}{2} (2J+1) \right. \\ \times [E^* - V(\mathbf{q}) - B_{\text{ext}(i)} J(J+1)] [E^* - V(\mathbf{q}) - A_{\text{ext}(i)} J(J+1)]^{1/2} + \frac{3}{(A_{\text{ext}(i)} - B_{\text{ext}(i)})^{1/2}} \\ \times [E^* - V(\mathbf{q}) - B_{\text{ext}(i)} J(J+1)]^2 \sin^{-1} \left\{ \frac{(2J+1) (A_{\text{ext}(i)} - B_{\text{ext}(i)})^{1/2}}{2 [E^* - V(\mathbf{q}) - B_{\text{ext}(i)} J(J+1)]^{1/2}} \right\} \Bigg), \\ A_{\text{ext}(i)} J(J+1) \leq E^* - V(\mathbf{q}), \quad (44b)$$

where $B_{\text{ext}(i)}$ and $A_{\text{ext}(i)}$ represent the external rotational constants appropriate to the configurations corresponding to Eqs. (43a)–(43e). In the evaluation of Eqs. (44a) and (44b) the overall flexible transition state has been approximated as a prolate top. This is almost always a good approximation, because of the extreme prolateness of flexible transition states. The external rotational constants $B_{\text{ext}(i)}$ and $A_{\text{ext}(i)}$ are therefore given by

$$B_{\text{ext}(1)} = B_{012}, \quad A_{\text{ext}(1)} = \infty, \quad (45a)$$

$$B_{\text{ext}(2)} = (B_{01}B_{012})^{1/2}, \quad A_{\text{ext}(2)} = B_2, \quad (45b)$$

$$B_{\text{ext}(3)} = (B_{02}B_{012})^{1/2}, \quad A_{\text{ext}(3)} = B_1, \quad (45c)$$

$$B_{\text{ext}(4)} = (B_0B_{012})^{1/2}, \quad A_{\text{ext}(4)} = B_{12}, \quad (45d)$$

$$B_{\text{ext}(5)} = (B_{02}B_{01})^{1/2}, \quad A_{\text{ext}(5)} = B_{12}, \quad (45e)$$

In Fig. 5, the results of Eqs. (44a) and (44b) are illustrated graphically for a representative system containing two HCN molecules ($B = 1.48 \text{ cm}^{-1}$) at a separation of 4.5 Å. To avoid confusion, just three of the five configurations of Eqs. (43a)–(43e) are included: those of Eq. (43a) (curve I), Eq. (43c) (curve II), and Eq. (43e) (curve III) [for two identical fragments, Eqs. (43b) and (43c) are the same]. It is apparent that, for the case of two linear fragments, the momentum-space volume is very sensitive to the relative orientation of the fragments at low- J values, favoring a linear configuration as compared with a nonlinear one. Put in simple physical terms, this is because the linear configuration has *four* internal momenta amongst which the energy $E^* - V(\mathbf{q}) - B_{\text{ext}}J(J+1)$ may be distributed in any fashion. A nonlinear configuration, however, can distribute this energy among only three internal momenta plus the external “ K rotor,” which is restricted by the constraint $-J \leq K \leq J$. The additional restriction on the K rotor of the nonlinear configuration reduces the number of momentum states available as compared with the linear configuration. This difference becomes negligible at higher J values, since the energy $E^* - V(\mathbf{q}) - B_{\text{ext}}J(J+1)$ is then small enough that the limits on the K rotor of the nonlinear configurations are determined energetically and not by the total angular momentum J . Note that all three curves of Fig. 5 are evaluated for the same available kinetic energy $E^* - V(\mathbf{q}) = 1000 \text{ cm}^{-1}$, hence the effect of the potential $V(r) + V(\mathbf{q})$ on the available momentum space has been separated out and the differences observed are related purely to the angular momentum considerations outlined above.

Configurations in which the principal axes of the two fragments in the flexible transition state are not coaligned

onto a common Cartesian set of axes do not prove amenable to simple analytic evaluation of the available momentum space $\Phi(E, J, \mathbf{q})$ in the manner described above, as a result of coupling terms which arise between the internal and external momenta. However, the individual configurations for which one can easily evaluate $\Phi(E, J, \mathbf{q})$ cover the range of configurational distortions of the overall body that are possible by altering the orientations of the fragments. A straightforward approach is therefore to seek a simple functional form which interpolates correctly through each of these known functions as the configuration of the system is varied continuously. The method of choice is simply to allow the external rotational constants B_{ext} and A_{ext} to vary continuously as a function of the configuration $\mathbf{q}$. As is apparent from Eqs. (44a) and (44b) this gives the desired behavior.

The procedure for analyzing the configurational dependence of $\Phi(E, J, \mathbf{q})$ in a flexible transition state comprised of two linear fragments may be extended to linear + nonlinear and nonlinear + nonlinear fragment combinations in a straightforward manner. The results for $\Phi(E, J, \mathbf{q})$ for all relevant combinations of atom, linear, spherical-top, symmetric-top (both oblate and prolate), and asymmetric top fragments may be concisely summarized as follows. For high J :

$$\begin{aligned} \Phi(E, J, \mathbf{q}) &= \frac{B_{\text{ext}}(\mathbf{q}) [A_{\text{ext}}(\mathbf{q})]^{1/2}}{(2J+1)} \frac{B_0 [A_{\text{ext}}(\mathbf{q}) - B_{\text{ext}}(\mathbf{q})]^{1/2}}{\times \left[\prod_{i=1}^S \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \frac{\Gamma(1/2)^{s-p}}{\Gamma(s/2+1)}} \\ &\times [E^* - V(\mathbf{q}) - B_{\text{ext}}(\mathbf{q})J(J+1)]^{s/2}, \end{aligned}$$

$$B_{\text{ext}}(\mathbf{q})J(J+1) \leq E^* - V(\mathbf{q}) \leq A_{\text{ext}}(\mathbf{q})J(J+1), \quad (46a)$$

where s is the total number of fragment rotational degrees of freedom (e.g., for a linear + linear system, $s = 4$; for a nonlinear + nonlinear system, $s = 6$), $\Gamma(x)$ is the gamma function, and the product is over all principal-axis rotational constants Λ_i of the fragments (degenerate rotational constants, such as for a linear fragment or a spherical top, being repeated the appropriate number of times). The integer $p = 2$ if one of the fragments is monatomic, otherwise $p = 4$. For low J :

$$\begin{aligned} \frac{\Phi(E, J, \mathbf{q})}{(2J+1)} &= \frac{B_{\text{ext}}(\mathbf{q}) [A_{\text{ext}}(\mathbf{q})]^{1/2}}{B_0} \left[\prod_{i=1}^S \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \\ &\times \frac{\Gamma(1/2)^{s-p-1}}{\Gamma((s+1)/2)} G(E, J, \mathbf{q}), \\ A_{\text{ext}}(\mathbf{q})J(J+1) &\leq E^* - V(\mathbf{q}), \end{aligned} \quad (46b)$$

where $G(E, J, \mathbf{q})$ has the following forms for even and odd s values, respectively:

$$\begin{aligned} G(E, J, \mathbf{q}) &= (2J+1) \left\{ \sum_{\nu=0}^{s/2-1} \frac{(s-1; -2; \nu)}{(s; -2; \nu+1)} [E^* - V(\mathbf{q}) - B_{\text{ext}}(\mathbf{q})J(J+1)]^\nu \right. \\ &\times [E^* - V(\mathbf{q}) - A_{\text{ext}}(\mathbf{q})J(J+1)]^{(s-1)/2-\nu} \Big\} \\ &+ \left(\frac{1}{2} \right)^{(s-2)/2} \frac{(1; 2; s/2)}{(s/2)!} \frac{[E^* - V(\mathbf{q}) - B_{\text{ext}}(\mathbf{q})J(J+1)]^{s/2}}{[A_{\text{ext}}(\mathbf{q}) - B_{\text{ext}}(\mathbf{q})]^{1/2}} \\ &\times \sin^{-1} \left\{ \frac{(2J+1) [A_{\text{ext}}(\mathbf{q}) - B_{\text{ext}}(\mathbf{q})]^{1/2}}{2 [E^* - V(\mathbf{q}) - B_{\text{ext}}(\mathbf{q})J(J+1)]^{1/2}} \right\} \quad \text{for } s \text{ even} \end{aligned} \quad (47a)$$

and

$$G(E, J, \mathbf{q}) = (2J + 1) \left\{ \sum_{\nu=0}^{(s-1)/2} \frac{(s-1; -2, \nu)}{(s; -2; \nu+1)} \right. \\ \times [E^* - V(\mathbf{q}) - B_{\text{ext}}(\mathbf{q})J(J+1)]^\nu \\ \times [E^* - V(\mathbf{q}) - A_{\text{ext}}(\mathbf{q})J(J+1)]^{(s-1)/2 - \nu} \left. \right\} \\ \text{for } s \text{ odd,} \quad (47b)$$

where the notation (x, n, ν) has the following meaning:

$$(x; n; \nu) = \prod_{j=1}^{\nu} [x + (j-1)n].$$

If there are any additional one-dimensional torsional rotors internal to the fragments of the flexible transition state, these may simply be included in the total number of rotational degrees of freedom s in Eqs. (46) and (47), and their corresponding rotational constants included in the product.

It should be stressed that in Eqs. (46) and (47) the external rotational constants $A_{\text{ext}}(\mathbf{q})$ and $B_{\text{ext}}(\mathbf{q})$ are to be taken as functions of the configuration $\mathbf{q}$ since the inertial symmetry of the overall body changes with varying fragment configurations. As noted above, this provides a convenient interpolation of the functional form of $\Phi(E, J, \mathbf{q})$ between the known functions $\Phi(E, J, \mathbf{q}_i)$ for those configurations in which the fragment principal axes are coaligned onto a common set of Cartesian axes.

Typical results for two representative systems are illustrated in Fig. 6: the $\text{CH}_3 + \text{CH}_3$ (oblate top + oblate top) system in Fig. 6(a) and the $\text{CH}_2\text{Cl} + \text{CH}_2\text{Cl}$ (asymmetric top + asymmetric top) system in Fig. 6(b). In each case, $\Phi(E, J, \mathbf{q})$ is plotted for two limiting configurations which represent the extreme changes in the inertial symmetry of the overall body with respect to changing fragment orientation. Curve I is for the configuration in which the "unique" principal axis (i.e., that with a rotational constant distinctly separated from the other two) of each fragment lies along the line joining the centers of mass. Curve II is for the configuration in which the "unique" principal axis of each fragment and the line between the centers of mass are all mutually orthogonal (the CH_2Cl case is additionally specified by the principal axis with the smallest rotational constant being oriented along the line joining the centers of mass). The available kinetic energy and interfragment separations are the same as for the earlier figures. A comparison of Fig. 6(a) for the methyl + methyl flexible transition state with curve I of Fig. 1(a) indicates the extent of error introduced by approximating a symmetric top for which the rotational constant A and B differ by a factor of 2 ($B_{\text{CH}_3} = 9.45 \text{ cm}^{-1}$, $A_{\text{CH}_3} = 4.73 \text{ cm}^{-1}$) as a spherical top. In this case, the angular-momentum-induced configurational dependence of $\Phi(E, J, \mathbf{q})$ is not great, indicating that for these purposes CH_3 is "almost a spherical top." In comparison, the $\text{CH}_2\text{Cl} + \text{CH}_2\text{Cl}$ system displays a much greater angular-momentum-induced configurational dependence in $\Phi(E, J, \mathbf{q})$ at low- J values. This also is to be expected, since the CH_2Cl radical is highly prolate (using stan-

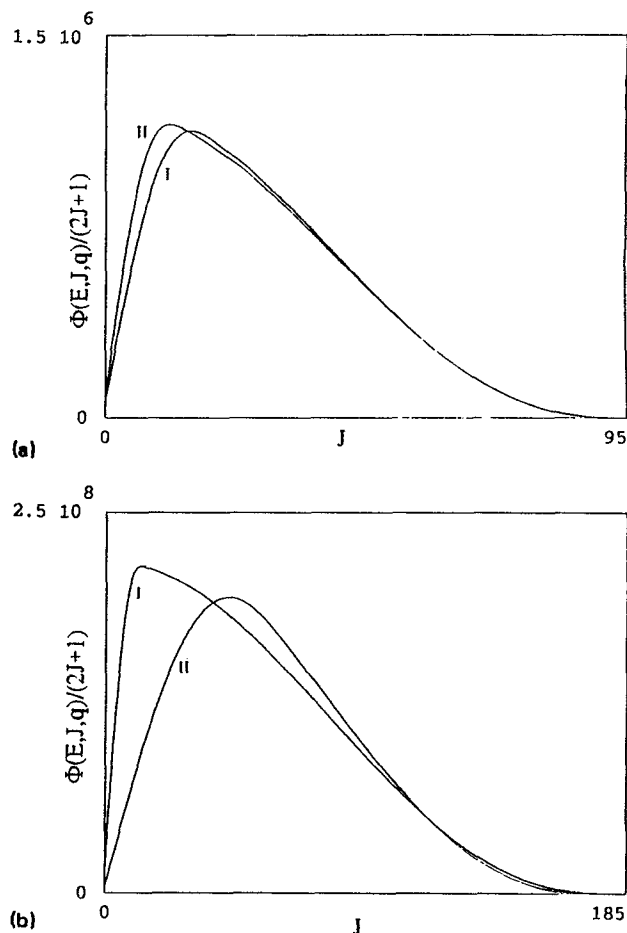


FIG. 6. Plot of $\Phi(E, J, \mathbf{q})$ vs angular momentum J for (a) the $\text{CH}_3 + \text{CH}_3$ system, and (b) the $\text{CH}_2\text{Cl} + \text{CH}_2\text{Cl}$ system. Center-of-mass separations $4.5 \text{ \AA}$. Available kinetic energy $E^* - V(\mathbf{q}) = 1000 \text{ cm}^{-1}$. CH_3 rotational constants: $A = 4.73 \text{ cm}^{-1}$, $B = 9.45 \text{ cm}^{-1}$. CH_2Cl rotational constants: $A = 9.45 \text{ cm}^{-1}$, $B = 0.48 \text{ cm}^{-1}$, $C = 0.46 \text{ cm}^{-1}$. Curve I: "unique" principal axes of the fragments aligned along the axis joining the centers of mass. Curve II: unique principal axes and axis joining the centers of mass all mutually perpendicular (principal axes of CH_2Cl fragments with smallest rotational constants aligned along the axis joining centers of mass).

dard bond lengths and bond angles, $A_{\text{CH}_2\text{Cl}} = 9.45 \text{ cm}^{-1}$, $B_{\text{CH}_2\text{Cl}} = 0.48 \text{ cm}^{-1}$, and $C_{\text{CH}_2\text{Cl}} = 0.46 \text{ cm}^{-1}$), and represents an intermediate case between nearly spherical fragments [Fig. 6(a)] and linear fragments (Fig. 5).

Equation (46) represents a substantial simplification of the overall problem of evaluating the transitional-mode sum of states $W_{\text{TM}}(E, J)$, since it completely removes the problem of integrating numerically over the momentum variables. The *only* approximation used in the evaluation of the individual $\Phi(E, J, \mathbf{q}_i)$ [for the linear + linear case, Eqs. (44) and (45)], upon which the general result is based, is that of treating the overall body as a prolate top. This is normally an excellent approximation for flexible transition states, wherein the fragments are separated typically by more than $3 \text{ \AA}$ and so the overall body is highly prolate, the rotational constants B_{ext} and C_{ext} for rotation about the principal axes perpendicular to the axis joining the centers of mass being almost equal. Indeed, this approximation is much less severe than that commonly used in classical phase-space theory.^{30(c),20} wherein the inertial asymmetry of *individual*

nonlinear fragments is neglected by treating them as spherical tops. In the present work, full account is taken of inertial asymmetry of the individual fragments. For arbitrary configurations $\mathbf{q}$, Eq. (46) may be viewed as an interpolation between the known functional forms $\Phi(E, J, \mathbf{q}_i)$ for all the reference configurations $\mathbf{q}_i$, wherein the principal axes of the fragments coalign onto a common Cartesian set of axes. Given that the effect on $\Phi(E, J, \mathbf{q})$ of inertial asymmetry in the fragments is generally only a small-to-moderate one (as indicated by Figs. 5 and 6), the interpolation is expected to be very accurate.

It is easily shown that, as the separation between the fragments becomes large and hence the orbital rotor dominates the density of states in the flexible transition state, $\Phi(E, J, \mathbf{q})$ approaches that predicted by the $J \approx L$ limit. As the separation becomes large, one has $B_{\text{ext}}(\mathbf{q}) \approx B_0 \gg A_{\text{ext}}(\mathbf{q})$. Hence the angular momentum $J^* \approx [E^* - V(\mathbf{q})]^{1/2} / A_{\text{ext}}(\mathbf{q})^{1/2}$ at which the transition between the low- and high- J forms of $\Phi(E, J, \mathbf{q})$ occurs becomes negligible compared with the overall range of allowable angular momenta, $0 < J < J_{\text{max}}$ where

$$J_{\text{max}} \approx [E^* - V(\mathbf{q})]^{1/2} / B_{\text{ext}}(\mathbf{q})^{1/2} \\ \approx [E^* - V(\mathbf{q})]^{1/2} / B_0^{1/2}.$$

Therefore, the low- J form of $\Phi(E, J, \mathbf{q})$ becomes irrelevant for large separations, and one need only examine the behavior of the high- J form, Eq. (46a). Hence we have

$$\lim_{J \rightarrow \infty} \frac{\Phi(E, J, \mathbf{q})}{(2J+1)} \approx \left[\prod_{i=1}^S \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \frac{\Gamma(1/2)^{s-p}}{\Gamma(s/2+1)} \\ \times [E^* - V(\mathbf{q}) - B_0 J(J+1)]^{s/2}, \quad (48)$$

which is equivalent to the sum of states for a collection of one- and two-dimensional free internal rotors that is used in the $J \approx L$ model in the limit of free rotors (see, e.g., Ref. 16).

In the next section, the evaluation of $W_{\text{TM}}(E, J)$ by integration of $\Phi(E, J, \mathbf{q})$ over the configuration space will be considered. For the development of accurate analytical approximations to $W_{\text{TM}}(E, J)$, it is necessary to simplify the functional form of $\Phi(E, J, \mathbf{q})$ at low- J values [Eqs. (46b) and (47)]. Analogous to the sphere + sphere case of Sec. II [Eqs. (38) and (39)], the appropriate low- J approximation to Eq. (46b) for both even and odd values of s is

$$\frac{\Phi(E, \text{low } J, \mathbf{q})}{(2J+1)} = (2J+1) \frac{B_{\text{ext}}(\mathbf{q}) [A_{\text{ext}}(\mathbf{q})]^{1/2}}{B_0} \\ \times \left[\prod_{i=1}^S \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \frac{\Gamma(1/2)^{s-p-1}}{\Gamma((s+1)/2)} \\ \times [E^* - V(\mathbf{q}) - B_{\text{av}} J(J+1)]^{(s-1)/2}, \quad (49)$$

where $B_{\text{av}} = (B_{\text{ext}}^2 A_{\text{ext}})^{1/3}$.

IV. EVALUATION OF THE TRANSITIONAL-MODE SUM OF STATES

Sections II and III have shown how one may derive accurate analytic expressions for the available momentum-space volume $\Phi(E, J, \mathbf{q})$ at a given configuration $\mathbf{q}$ of the fragments in the flexible transition state. It remains to inte-

grate $\Phi(E, J, \mathbf{q})$ over the configuration space $\mathbf{q}$ in order to obtain the transitional-mode sum of states $W_{\text{TM}}(E, J)$ [Eq. (15)]. This may be carried out to optimum accuracy by the numerical evaluation of Eq. (15). As will be shown below and in Appendix C this involves at most a five-dimensional numerical integral, since three angles may always be associated with the rotation of the overall body. For arbitrary interaction potentials, the numerical approach is clearly necessary. Although *ab initio* calculations of the interaction potential for arbitrary configurations at the range of separations involved in flexible transition states (see, e.g., Ref. 42) are still largely lacking, a number of model potential forms have been suggested, (e.g., Refs. 35, 36, 2, 40, 37, 26, 49, and 50). In the present work, we seek accurate analytic results for $W_{\text{TM}}(E, J)$ by the use of simple model potentials for $V(\mathbf{q})$ which describe the important features of the noncentral interaction potential between the two fragments, and have sufficient flexibility to enable fitting to more accurate potentials as they become available. This then enables reliable microcanonical variational calculations of $k(E, J)$ to be carried out rapidly and easily.

The first task encountered in evaluating Eq. (15) for $W_{\text{TM}}(E, J)$ is to transform to a set of angular coordinates comprised of "external" angles that involve rotation only of the overall body and "internal" angles which involve relative motion of the two fragments. The advantage of such a set of coordinates is that the external angles, since they do not involve a change in the orientation of the fragments relative to each other, will not appear in the interaction potential $V(\mathbf{q})$; they can therefore be integrated over immediately to reduce the problem to at most a five-dimensional integral. In the following, transformations appropriate to a nonlinear + nonlinear system will be outlined; the transformations for simpler systems then follow directly by setting the extraneous Euler angles of the fragments to zero. It should be noted that there is no canonically conjugate relationship between the angular coordinates defined below and the external and internal angular momenta of the previous sections: The coordinate transformations are simply a useful device for facilitating ease of evaluation of the configuration-space integral of Eq. (15), and the relevant Jacobian for the transformations must be evaluated. It is convenient to transform first to a set of Euler angles $(\phi'_i, \theta'_i, \psi'_i)$ for the fragments which are referenced relative to the principal axes of the orbital rotor rather than the absolute spatially fixed Cartesian axes:

$$\{\phi_i, \theta_i, \psi_i\} \xrightarrow{J_c = \sin \theta'_i / \sin \theta_i} \{\phi'_i, \theta'_i, \psi'_i\}, \quad (50)$$

where the new Euler angles are related to the old by the relations

$$\cos \theta'_i = \sin \theta \sin \theta_i (\sin \phi \sin \phi_i + \cos \phi \cos \phi_i) \\ + \cos \theta \cos \theta_i, \quad (51a)$$

$$\sin \phi'_i = \frac{\sin \theta_i}{\sin \theta'_i} (\cos \phi \sin \phi_i - \sin \phi \cos \phi_i), \quad (51b)$$

$$\psi'_i = \phi_i + \psi_i - \phi - \phi'_i. \quad (51c)$$

Equations (51a)–(51c) and the corresponding Jacobian are

derived in Appendix C. These transformation equations are derived for the case where the fragment-fixed axes are the principal axes of the fragment. However, the angles $(\theta'_i, \phi'_i, \psi'_i)$ may in fact be regarded as angles which relate the orientation of *any* fragment-fixed Cartesian set of axes, with origin located at the fragment center of mass, to the orbital principal axes. This does not affect the form of the integral for the sum of states $W_{\text{TM}}(E, J)$. The important consideration is that the angular coordinates chosen must properly span the configuration space, and a rotation of Cartesian axes to obtain a new set of fragment-fixed axes will always satisfy this criterion. This fact will be made use of below in the evaluation of $W_{\text{TM}}(E, J)$, where it is often useful to choose fragment-fixed axes which take advantage of the symmetry in the bonding potential rather than the inertial symmetry of the fragment. This flexibility in the choice of angular coordinates arises from the fact that noncanonical transformations were used to evaluate the available momentum-space volume $\Phi(E, J, \mathbf{q})$: one is not tied by the requirement of a canonical relationship between coordinates and momenta.

Next, one transforms from the two ϕ'_i coordinates for the independent rotation of each fragment about the orbital rotor z axis (i.e., the line joining the centers of mass) to an angle ψ for rotation of the overall body and a torsional angle ϕ_t which describes the relative twisting of the two fragments about the axis before the independent (θ'_1, ψ'_1) and (θ'_2, ψ'_2) rotations are carried out:

$$\{\phi'_1, \phi'_2\} \xrightarrow{J_c=1} \{\psi, \phi_t\}, \quad (52)$$

where

$$\psi = \phi'_1, \quad (53a)$$

$$\phi_t = \phi'_2 - \phi'_1. \quad (53b)$$

The complete set of coordinates is now $\{\phi, \theta, \psi, \phi_t, \theta'_1, \psi'_1, \theta'_2, \psi'_2\}$. In order to clarify the meaning of this set of coordinates, consider the sequence of rotations which takes the system to an arbitrary configuration from a reference configuration in which the principal axes of both fragments are coaligned with the spatially fixed Cartesian axes, and the orbital rotor z axis (i.e., the line joining the centers of mass) is aligned along the spatially fixed z axis:

(i) The overall system is rotated as a rigid body through the orbital-rotor Euler angles (ϕ, θ) so that the orbital rotor z axis is correctly aligned. The fragments would now require rotations through the relative Euler angles $(\phi'_1, \theta'_1, \psi'_1)$ and $(\phi'_2, \theta'_2, \psi'_2)$, respectively, in order to move into their correct orientations.

(ii) The overall system is rotated as a rigid body through the angle $\psi = \phi'_1$ about the line joining the centers of mass, so that fragment 1 is ready for its second (θ'_1) Euler-angle rotation.

(iii) Fragment 2 is rotated through the angle $\phi_t = \phi'_2 - \phi'_1$ about the line joining the centers of mass (fragment 1 being held frozen) so that it is also ready for its second (θ'_2) Euler-angle rotation.

(iv) Both fragments are now rotated independently through

their respective (θ'_i, ψ'_i) Euler angles so as to bring them into the required configurations.

The sum of states is written in terms of these coordinates as

$$W_{\text{TM}}(E, J) = \frac{1}{\sigma_1 \sigma_2} \frac{1}{N} \int_0^{2\pi} d\phi \int_0^{2\pi} d\psi \int_0^\pi d\theta \sin \theta \\ \times \int_0^{2\pi} d\phi_t \int_0^{2\pi} d\psi'_1 \int_0^{2\pi} d\psi'_2 \int_0^\pi d\theta'_1 \\ \times \sin \theta'_1 \int_0^\pi d\theta'_2 \sin \theta'_2 \Phi(E, J, \mathbf{q}). \quad (54)$$

Now, since steps (i) and (ii) above do not affect the potential $V(\mathbf{q})$ or the external moments of inertia $A_{\text{ext}}(\mathbf{q})$ and $B_{\text{ext}}(\mathbf{q})$ of the system, it is apparent that $\Phi(E, J, \mathbf{q})$ [Eqs. (46) and (47)] is independent of the “external” angles ϕ , θ , and ψ . Hence these angles may be integrated over to give

$$W_{\text{TM}}(E, J) = \frac{1}{\sigma_1 \sigma_2} \frac{1}{2^5 \pi^3} \int_0^{2\pi} d\phi_t \int_0^{2\pi} d\psi'_1 \int_0^{2\pi} d\psi'_2 \int_0^\pi d\theta'_1 \\ \times \sin \theta'_1 \int_0^\pi d\theta'_2 \sin \theta'_2 \Phi(E, J, \mathbf{q}). \quad (55)$$

Evaluation of $W_{\text{TM}}(E, J)$ for an arbitrary interaction potential $V(\mathbf{q})$ therefore involves at most a five-dimensional numerical integral.

Equation (55), with $\Phi(E, J, \mathbf{q})$ therein given by Eq. (46) of the preceding section, is the most accurate expression for the transitional-mode sum of states $W_{\text{TM}}(E, J)$ presented in this work. It is classically exact for a spherical-top-spherical-top flexible transition state. For inertially asymmetric fragments, Eq. (55) may be regarded as a highly accurate approximate solution. The nature of the approximations involving asymmetric fragments, and the reasons for their accuracy, are as summarized in the discussion between Eqs. (47) and (48) of the preceding section. The evaluation of Eq. (55), while considerably faster than the exact numerical methods (e.g., Refs. 4 and 23) due to the reduced dimensionality of the integral, requires nevertheless significant computing time. For the purposes of modeling pressure- and temperature-dependent falloff data, where the required accuracy is not as great as for detailed state-specific experiments and many repetitive calculations spanning a wide range of different conditions are necessary, a certain amount of accuracy may be sacrificed for the sake of speed and convenience. In what follows we therefore introduce additional approximations, each discussed in turn, that lead to analytic expressions for $W_{\text{TM}}(E, J)$. These expressions, while not as accurate as the numerical evaluation of Eq. (55), are nevertheless substantially more accurate than the existing simpler models (e.g., Refs. 16, 26, 37, and 42), and facilitate extremely rapid implementation of variational RRKM theory.

The further reduction of Eq. (55) with a view to obtaining accurate analytic approximations to $W_{\text{TM}}(E, J)$ requires approximations to be made concerning (1) the low- J form of the momentum-space volume $\Phi(E, J, \mathbf{q})$, (2) the configurational dependence of $\Phi(E, J, \mathbf{q})$ that is induced by inertial asymmetry in the fragments (recall that this is independent of the configurational dependence brought about by the non-

central potential $V(\mathbf{q})$], and (3) the use of simplified model forms of the potential $V(\mathbf{q})$. These approximations are now discussed in turn.

The necessary simplified expressions for the low- J form of $\Phi(E, J, \mathbf{q})$ have been introduced in Secs. II and III [Eq. (38) for the spherical-top-spherical-top system, and Eq. (49) for the general case]. This is generally a very accurate approximation as indicated in Figs. 3(a)–3(c).

The configurational dependence of the available momentum-space volume induced by inertial asymmetry in the fragments has been illustrated in Sec. III (Figs. 5 and 6). This dependence, which is commonly neglected in the application of RRKM theory, is encapsulated in Eqs. (46) and (47) through the configurational dependence of the external rotational constants: $A_{\text{ext}}(\mathbf{q})$ and $B_{\text{ext}}(\mathbf{q})$. The dependences of A_{ext} and B_{ext} on $\mathbf{q}$ are complex for all but the simplest systems, and hence Eq. (55) cannot be evaluated analytically without recourse to approximation. That a suitable approximation is possible becomes apparent by further consideration of the representation of the sum of states $W_{\text{TM}}(E, J)$ as an average value of $\Phi(E, J, \mathbf{q})$, Eq. (15). One notes that, at large separations, the hindrance to the rotation of the fragments is negligible: they are essentially freely rotating products. In this limit, the sum of states $W_{\text{TM}}(E, J)$ becomes equivalent to the average of $\Phi(E, J, \mathbf{q})$ for a central potential, given by $W_{\text{CP}}(E, J)$ in Eq. (40). However, as noted in the preceding section, at large separations Eq. (46) [and, indeed, Eq. (40)] approaches the $J \approx L$ limit [Eq. (48)] and so the configurational dependence of $A_{\text{ext}}(\mathbf{q})$ and $B_{\text{ext}}(\mathbf{q})$ becomes unimportant. At smaller separations, the noncentral potential $V(\mathbf{q})$ weights the average strongly in favor of the minimum-energy configuration, i.e., the reaction coordinate configuration $\mathbf{q}_s$, since the available kinetic energy $E^* - V(\mathbf{q}) = E - V(r) - V(\mathbf{q})$ is a maximum for this configuration. In order to avoid the complexities introduced by the dependence of the external rotational constants on the fragment configurations, $A_{\text{ext}}(\mathbf{q})$ and $B_{\text{ext}}(\mathbf{q})$ are therefore approximated as $A_{\text{ext}}(\mathbf{q}_s)$ and $B_{\text{ext}}(\mathbf{q}_s)$, respectively. This approximation is accurate at smaller separations because of the strong weighting in favor of the minimum-energy configuration $\mathbf{q}_s$ induced by the potential $V(\mathbf{q})$ when there is significant bonding, and goes to the correct limit at large separations because the configurational dependence $A_{\text{ext}}(\mathbf{q})$ and $B_{\text{ext}}(\mathbf{q})$ is irrelevant at large separations. The high- and approximate low- J forms of $\Phi(E, J, \mathbf{q})$ may then be written in the general form:

$$\frac{\Phi(E, J, \mathbf{q})}{(2J + 1)} = C [\xi - V(\mathbf{q})]^m, \quad (56)$$

where, neglecting the configurational dependence of $A_{\text{ext}}(\mathbf{q})$ and $B_{\text{ext}}(\mathbf{q})$ as discussed above, the quantities in Eq. (56) are as follows.

For high J [Eq. (46a)]:

$$C = \frac{B_{\text{ext}}(\mathbf{q}_s) [A_{\text{ext}}(\mathbf{q}_s)]^{1/2}}{B_0 [A_{\text{ext}}(\mathbf{q}_s) - B_{\text{ext}}(\mathbf{q}_s)]^{1/2}} \left[\prod_{i=1}^s \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \times \frac{\Gamma(1/2)^{s-p-1}}{\Gamma(s/2 + 1)}, \quad (57a)$$

$$\xi = E^* - B_{\text{ext}}(\mathbf{q}_s) J(J + 1), \quad (57b)$$

$$m = s/2. \quad (57c)$$

For low J [Eq. (49)]:

$$C = (2J + 1) \frac{B_{\text{ext}}(\mathbf{q}_s) [A_{\text{ext}}(\mathbf{q}_s)]^{1/2}}{B_0} \left[\prod_{i=1}^s \left(\frac{1}{\Lambda_i} \right)^{1/2} \right] \times \frac{\Gamma(1/2)^{s-p-1}}{\Gamma(s + 1/2)}, \quad (57d)$$

$$\xi = E^* - B_{\text{av}} J(J + 1), \quad (57e)$$

$$m = (s - 1)/2. \quad (57f)$$

The evaluation of Eq. (55) for $W_{\text{TM}}(E, J)$, with $\Phi(E, J, \mathbf{q})$ therein as specified in Eqs. (56) and (57), will now be considered. The total interaction potential V is written as a sum of the reaction coordinate potential $V(s)$, i.e., the minimum-energy pathway, and the potential $V(\mathbf{q})$ for the degrees of freedom orthogonal to the reaction coordinate:

$$V = V(s) + V(\mathbf{q}; s), \quad (58)$$

where $V(\mathbf{q})$ has been written explicitly in Eq. (58) as a function which varies parametrically with the reaction coordinate s . It is common when modeling systems with flexible transition states to select as the reaction coordinate the separation r between the fragment centers of mass. This is appropriate because of the typically large (i.e., ≥ 3 Å) separation between the fragments, and has the advantage of a comparatively simple expression for the kinetic energy [Eq. (6)]. $V(\mathbf{q})$ is commonly written as a sum of bonding and nonbonding contributions (e.g., Ref. 4):

$$V(\mathbf{q}) = V_b(\mathbf{q}) + V_{\text{nb}}(\mathbf{q}), \quad (59)$$

where $V_b(\mathbf{q})$ represents the *angular* dependence of the bonding potential [the dependence on the separation of the fragments is implicit in $V(r)$] and $V_{\text{nb}}(\mathbf{q})$ represents the sum of the nonbonding interaction potentials (i.e., steric interactions between the fragments). Nonbonding interactions are represented crudely in the simple Gorin model (see, e.g., Ref. 51) by treating the nonbonded atoms as hard spheres, which results in a certain fraction of the configuration space being excluded. Nonbonded interactions effectively exclude configurations involving overlap of nonbonded atoms because the steep repulsive potentials between such atoms make these configurations energetically inaccessible. However, the hard-sphere approximation appears to be too extreme at low energies, and can lead to discontinuous behavior in the sum of states $W_{\text{TM}}(E, J)$ as a function of the separation distance r .^{52,26} A more accurate way of treating the nonbonded interactions is to use Lennard-Jones interaction potentials for all of the pairwise nonbonding interactions between atoms from different fragments.⁴ In this case, evaluation of Eq. (55) for $W_{\text{TM}}(E, J)$ requires numerical integration. Clearly, at lower energies the nonbonded interactions are “softer” than hard spheres. Under these conditions, the Gorin model is not appropriate and a model for $V(\mathbf{q})$ based on the bonding potential $V_b(\mathbf{q})$ (with possibly some adjustment for “soft” steric interactions) is more valid.²⁶ However, as the energy increases, the steep repulsive walls of the nonbonded interactions become the most relevant part of these potentials, and the hard-sphere approxi-

mation may be expected to be more valid. It is therefore useful to evaluate Eq. (55) with two limiting potentials, the first being a central potential (i.e., no angular dependence) with hard-sphere nonbonded interactions between the fragments, and the second being a model noncentral potential which incorporates the main features of the angular dependence of the bonding potential V_b .

The central potential with hard-sphere nonbonded interactions is such that $V(\mathbf{q}) = 0$ except where there is some overlap of nonbonding atoms, in which case $V(\mathbf{q}) = \infty$. The condition for "overlap" of nonbonding atoms is that the separation between the atoms is less than an appropriately chosen hard-sphere collision diameter. Writing the sum of states for a central potential with hard-sphere nonbonded interactions as $W_{\text{TM}}^{\text{HS}}(E, J)$, Eq. (55) becomes for this case

$$\begin{aligned} \frac{W_{\text{TM}}^{\text{HS}}(E, J)}{(2J+1)} &= \frac{1}{\sigma_1 \sigma_2} \frac{C \xi^m}{2^5 \pi^3} \int_0^{2\pi} d\phi_1 \int_0^{2\pi} d\psi'_1 \int_0^{2\pi} d\psi'_2 \\ &\times \int_0^\pi d\theta'_1 \sin \theta'_1 \int_0^\pi d\theta'_2 \sin \theta'_2 \\ &\times \prod_{ij} S[r_{ij} - d_{ij}^{\text{HS}}], \end{aligned} \quad (60)$$

where r_{ij} is the separation between the i th atom in fragment 1 and the j th atom in fragment 2, and d_{ij}^{HS} is the mutual hard-sphere collision diameter assigned to the given pair of atoms (d_{ij}^{HS} should be chosen to approximate the steep repulsive wall of the Lennard-Jones interaction potential between the atoms). The step function $S[r_{ij} - d_{ij}^{\text{HS}}]$ takes the value of zero if r_{ij} becomes less than the assigned hard-sphere collision diameter d_{ij}^{HS} ; otherwise, $S[r_{ij} - d_{ij}^{\text{HS}}]$ is unity. The product of the step functions is for all pairwise nonbonded interactions between the fragments, so that if any two nonbonding atoms on different fragments overlap, the corresponding step function will take the value of zero and exclude the given configuration from the integral. For configurations where there is no steric overlap, $V(\mathbf{q}) \equiv 0$ and so ξ^m may be removed outside the integral. The angular integrations may be evaluated to yield

$$\frac{W_{\text{TM}}^{\text{HS}}(E, J)}{(2J+1)} = S[\xi] + \frac{C \xi^m \Delta \Omega}{\sigma_1 \sigma_2}, \quad (61)$$

where $\Delta \Omega$ is the fraction of the configuration space that is available to the system. In Eq. (61), the step function $S[\xi]$ has been introduced to ensure that the classically evaluated sum of states $W_{\text{TM}}(E, J)$ goes to the correct limit of one as $\xi \rightarrow 0$ (see, e.g., Ref. 20). Note that, since no hindrance to the fragment rotations is included except that of the hard-sphere nonbonded interactions, provided the d_{ij}^{HS} are estimated as reasonable *minimum* separations of the atoms, Eq. (61) represents an *upper bound* to the sum of states that would be calculated with inclusion of hindrance to the fragment rotations due to the bonding potential $V_b(\mathbf{q})$. This fact will be further utilized below.

The evaluation of $W_{\text{TM}}(E, J)$ for model potentials which represent just the bonding interaction $V_b(\mathbf{q})$ will now be considered. As noted above, a number of models of varying complexity have been proposed for the dependence of the

bonding potential on the orientation of the fragments. It is at least apparent that the potential is likely to be anharmonic and oscillatory in the angle of orientation, with a barrier height $V_0(r)$ to rotation of the fragments which varies in some way with the separation of the fragments (e.g., Ref. 31). In the models discussed below, the potential $V_b(\mathbf{q})$ is expressed as a function of Euler angles which specify the orientation of the fragments relative to the line joining the centers of mass (the orbital z axis). These angles relate the orientation of Cartesian axes fixed in the fragments, with origins located at the centers of mass, to the orientation of the orbital principal axes. For reactions wherein the bonding atoms of the fragments are *not* located at their centers of mass, it may be intuitively more useful to express the potential first as a function of angles and distances related to the bonding atoms rather than the centers of mass (e.g., Refs. 4, 23, 50, and 53). As described in the introduction, one then uses appropriate coordinate transformation equations to evaluate the potential in terms of the coordinates used in the actual calculation. In the present work, this additional transformation step is omitted. Even in the optimum situation where accurate *ab initio* data are available for the potential $V_b(\mathbf{q})$, one is faced with the task of fitting a potential surface with a suitable analytic expression, regardless of which coordinate system one chooses to start with. The approach presently adopted is to seek model potentials which enable rapid analytical evaluation $W_{\text{TM}}(E, J)$ and yet allow sufficient flexibility to model the essential features of accurate surfaces as they become available.

One further preliminary point deserves mention. In the development of the expressions for the available momentum-space volume $\Phi(E, J, \mathbf{q})$ and $W_{\text{TM}}^{\text{HS}}(E, J)$ above, the fragment-fixed Cartesian axial system has been that of the principal axes of the given fragment. In the evaluation of the configuration-space integral of Eq. (54) where the bonding potential $V_b(\mathbf{q})$, rather than steric effects, plays the dominant role in determining the sum of states; however, it is convenient to consider the fragment-fixed axes as being fixed in the fragment in such a way that the bonding potential may be accurately represented by the functional form chosen. As noted earlier in this section, this does not alter the form of the integral for $W_{\text{TM}}(E, J)$. One can therefore proceed without further complication with the understanding that the fragment-fixed axes have been chosen so that the bonding potential is well represented in terms of the model functional form chosen.

Recent work^{33,26} has shown that a sinusoidal hindrance to the fragment rotations is readily handled by classical phase-space integrations, yielding simple, exact analytical formulas for convoluted rotational densities or sums of states. The model for independent sinusoidally hindered rotors was recently presented together with detailed modeling calculations.²⁶ The work was carried out within the limits of the $J \approx L$ approximation, which treats the fragment rotations as independent and internal (i.e., having no contribution to the total angular momentum J) in the Hamiltonian kinetic energy (Sec. II), and also assumes that the fragments rotate under independent potentials. In Secs. II and III, a new method has been demonstrated for coupling the frag-

ment rotations with the orbital rotation in the Hamiltonian kinetic energy, so that their individual contributions to the total angular momentum J may be rigorously accounted for while retaining the considerable advantage of analytical expressions. The interaction potential between the fragments should also properly be coupled, since the interaction is perforce a mutual one and the potential which one fragment experiences will depend on its orientation *relative* to the other. It is also important, therefore, to utilize a coupled form for the interaction potential between the fragments, and to examine for a model case the error introduced into $W_{\text{TM}}(E, J)$ by the use of independent rotor potentials.

The models examined in this work involve independent and coupled sinusoidal hindrance to the fragment rotations. For simplicity, the coupled- and independent-rotor models will be compared only for the case of a flexible transition state with two symmetric radicals (i.e., radicals, such as CH_3 , with symmetric radical orbitals which have the potential for bond formation at orientations $\theta'_1 = 0$ and $\theta'_1 = \pi$). The coupled-rotor-potential formulas for $W_{\text{TM}}(E, J)$ for combinations of symmetric + asymmetric and asymmetric + asymmetric radicals will be presented at the end of this section.

Beginning with the assumption that the angular dependence of the bonding interaction potential, $V_b(\mathbf{q})$, can be well represented by a function of $\cos \theta'_i$:

$$V_b(\mathbf{q}) \approx V(\cos \theta'_1, \cos \theta'_2), \quad (62)$$

one may immediately evaluate the integrals over the cyclic coordinates in Eq. (55) to give

$$\frac{W_{\text{TM}}^b(E, J)}{(2J+1)} = \frac{C}{4\sigma_1\sigma_2} \int_0^\pi d\theta'_1 \sin \theta'_1 \int_0^\pi d\theta'_2 \sin \theta'_2 \times [\xi - V_b(\cos \theta'_1, \cos \theta'_2)]^m, \quad (63)$$

where the superscript b indicates that $W_{\text{TM}}(E, J)$ has been calculated on the basis of the model bonding potential $V_b(\mathbf{q})$, and does not include nonbonded steric interactions. The independent sinusoidally hindered rotor (ISHR) model of Jordan, Smith, and Gilbert²⁶ may be summarized for a

system with two symmetric fragment rotors as

$$V_b^{\text{ISHR}}(\mathbf{q}) = V_0(2 - |\cos \theta'_1| - |\cos \theta'_2|), \quad (64)$$

where V_0 is the barrier to rotation of either one of the fragments, and θ'_i is as defined above [Eq. (51)]. The barrier to rotation V_0 is a function of the separation r , and is commonly assumed to be associated with the energy required to break the partially formed bond, i.e., $V_0 = V(r = \infty) - V(r)$ (e.g., Refs. 2, 31, and 26). This reflects the fact that in rotating one of the fragments through an angle of $\pi/2$, one effectively removes the possibility of bonding overlap of the radical orbitals. A corresponding coupled sinusoidally hindered rotor (CSHR) model for the potential which admits analytical evaluation of the sum of states $W_{\text{TM}}(E, J)$ is

$$V_b^{\text{CSHR}}(\mathbf{q}) = V_0(1 - |\cos \theta'_1| |\cos \theta'_2|). \quad (65)$$

Note that both these functional forms have minima at $\theta'_i = 0$ and $\theta'_i = \pi$, because of the possibility of bond formation at both ends of the symmetric radicals. Somewhat different sinusoidal potentials have also been used by other workers.^{2,35,36} Substituting Eq. (64) into Eq. (63) and evaluating the ensuing integrations yields²⁶

$$\begin{aligned} \frac{W_{\text{TM}}^{\text{ISHR}}(E, J)}{(2J+1)} &= S[\xi] + \frac{C}{\sigma_1\sigma_2} \frac{1}{(m+1)(m+2)} \frac{1}{V_0^2} \\ &\times \{\xi^{m+2} - 2(\xi - V_0)^{m+2} S[\xi - V_0] \\ &+ (\xi - 2V_0)^{m+2} S[\xi - 2V_0]\}, \end{aligned} \quad (66)$$

where C , m , and ξ take values appropriate to the high- or low- J forms of $\Phi(E, J, \mathbf{q})$ as indicated by Eqs. (56) and (57), and $S[\xi]$ is included in order to obtain the correct threshold behavior as explained above. Substituting Eq. (65) into Eq. (63) and evaluating the ensuing integrals yields the following result:

$$\frac{W_{\text{TM}}^{\text{CSHR}}(E, J)}{(2J+1)} = S[\xi] + \frac{C}{\sigma_1\sigma_2} H_b(\xi), \quad (67)$$

where, for integral values of m , $H_b(\xi)$ (the subscript is for later notational convenience) is given by

$$\begin{aligned} H_b(\xi) &= \frac{1}{(m+1)} \left\{ \left[\frac{1}{(m+1)} V_0^m - \frac{(m+1)}{m} V_0^{m-1} (V_0 - \xi) + \frac{(m+1)m}{2!(m-1)} V_0^{m-2} (V_0 - \xi)^2 - \dots \right. \right. \\ &\quad \left. \left. + \frac{(m+1)m \dots 3}{(m-1)!2} V_0 (V_0 - \xi)^{m-1} - \frac{(m+1)!}{m!} (V_0 - \xi)^m \right] \right. \\ &\quad \left. - \frac{(V_0 - \xi)^{m+1}}{V_0} \left[\ln \left(\frac{V_0 - \xi}{V_0} \right) + \frac{1}{(m+1)} - \frac{(m+1)}{m} + \frac{(m+1)m}{2!(m-1)} - \dots \right. \right. \\ &\quad \left. \left. + \frac{(m+1)m \dots 3}{(m-1)!2} - \frac{(m+1)!}{m!} \right] \right\}, \quad \xi < V_0, \end{aligned} \quad (68a)$$

$$\begin{aligned} H_b(\xi) &= \frac{C}{\sigma_1\sigma_2} \left[(\xi - V_0)^m + \frac{m}{2 \cdot 2!} (\xi - V_0)^{m-1} V_0 + \frac{m(m-1)}{3 \cdot 3!} (\xi - V_0)^{m-2} V_0^2 + \dots \right. \\ &\quad \left. + \frac{1}{m} (\xi - V_0) V_0^{m-1} + \frac{1}{(m+1)^2} V_0^m \right], \quad \xi > V_0. \end{aligned} \quad (68b)$$

For half-integral values of m , $H_b(\xi)$ is given by

$$H_b(\xi) = \frac{C}{\sigma_1 \sigma_2} \frac{1}{(m+1)} \left[\left[\frac{1}{(m+1)} V_0^m - \frac{(m+1)}{m} V_0^{m-1} (V_0 - \xi) + \frac{(m+1)m}{2!(m-1)} V_0^{m-2} (V_0 - \xi)^2 - \frac{(m+1)m(m-1)}{3!(m-2)} V_0^{m-3} (V_0 - \xi)^3 + \dots \right] - \frac{(V_0 - \xi)^{m+1}}{V_0} \left[\frac{1}{(m+1)} - \frac{(m+1)}{m} + \frac{(m+1)m}{2!(m-1)} - \frac{(m+1)m(m-1)}{3!(m-2)} + \dots \right] \right], \quad \xi \leq V_0, \quad (68c)$$

$$H_b(\xi) = \frac{C}{\sigma_1 \sigma_2} \left[\frac{1}{(m+1)} \frac{(\xi - V_0)^{m+1}}{V_0} \ln \left(\frac{\xi - V_0}{V_0} \right) + \frac{(\xi - V_0)^{m+1}}{V_0} \left[1 + \frac{m}{2 \cdot 2!} + \frac{m(m-1)}{3 \cdot 3!} + \frac{m(m-1)(m-2)}{4 \cdot 4!} + \dots \right] - \frac{1}{(m+1)} \frac{(\xi - V_0)^{m+1}}{V_0} \left[\frac{1}{(m+1)} + \frac{(m+1)}{m} + \frac{(m+1)m}{2!(m-1)} + \frac{(m+1)m(m-1)}{3!(m-2)} + \dots \right] + \frac{1}{(m+1)} \left[\frac{1}{(m+1)} V_0^m + \frac{(m+1)}{m} V_0^{m-1} (\xi - V_0) + \frac{(m+1)m}{2!(m-1)} V_0^{m-2} (\xi - V_0)^2 + \frac{(m+1)m(m-1)}{3!(m-2)} V_0^{m-3} (\xi - V_0)^3 + \dots \right] \right], \quad V_0 < \xi \leq 2V_0, \quad (68d)$$

$$H_b(\xi) = \frac{C}{\sigma_1 \sigma_2} \left[(\xi - V_0)^m + \frac{m}{2 \cdot 2!} (\xi - V_0)^{m-1} V_0 + \frac{m(m-1)}{3 \cdot 3!} (\xi - V_0)^{m-2} V_0^2 + \frac{m(m-1)(m-2)}{4 \cdot 4!} (\xi - V_0)^{m-3} V_0^3 + \dots \right], \quad \xi \geq 2V_0. \quad (68e)$$

C , ξ , and m in Eqs. (67) and (68) are once again specified, appropriate to the high- or low- J form of $\Phi(E, J, \mathbf{q})$, in Eq. (57). In determining the results of Eqs. (67) and (68), the first integration over θ'_2 in Eq. (63) has been carried out in closed form and the second integral over θ'_1 has been carried out by straightforward series expansion. It is interesting to note that, using the method of this work, analytical results can also be obtained for the closely related coupled-rotor functional form for $V_b(\mathbf{q})$ used by Wardlaw and Marcus.² This function involves the product $\cos^2 \theta'_1 \cos^2 \theta'_2$ instead of the $|\cos \theta'_1| |\cos \theta'_2|$ that is used in Eq. (65). In addition, one can evaluate analytical expressions for $W_{\text{TM}}(E, J)$ for independent $\cos^2 \theta'_1$ hindering potentials. This provides flexibility in fitting the essential features of accurate *ab initio* or semiempirical potentials, while maintaining the advantage of very rapid evaluation of $W_{\text{TM}}(E, J)$. In order to avoid a confusion of analytical formulas, the results for the alternative potentials are not presented in this work. Instead, we focus on a comparison between the results for coupled and uncoupled potentials based on a common functional form for the rotational hindrance [Eqs. (64) and (65)]. Indeed, there is little advantage in choosing one functional form over another until sufficient *ab initio* results become available to indicate which is best adapted to fit the detailed form of $V_b(\mathbf{q})$.

Figure 7 illustrates results for the sum of states $W_{\text{TM}}(E, J)$ for the methyl + methyl system using the coupled-rotor potential (labeled I), the independent-rotor potential (labeled II), and, finally, the central potential (labeled III). The inertial parameters are the same as for Fig. 6(a). The high- and low- J forms of $W_{\text{TM}}^b(E, J)$ are plotted as a function of J for three different available kinetic ener-

gies: $E^* = 350 \text{ cm}^{-1}$ [4.2 kJ mol⁻¹; Fig. 7(a)], 1500 cm⁻¹ [17.9 kJ mol⁻¹; Fig. 7(b)], and 4000 cm⁻¹ [47.9 kJ mol⁻¹; Fig. 7(c)]. Consistent with Eq. (39) for the momentum-space volume, the sum of states $W_{\text{TM}}(E, J)$ should be chosen as the minimum of the high- and low- J forms:

$$W_{\text{TM}}(E, J) = \min\{W_{\text{TM}}(E, \text{low } J), W_{\text{TM}}(E, \text{high } J)\}. \quad (69)$$

It is apparent from Fig. 7 that the sum of states $W_{\text{TM}}^{\text{ISHR}}(E, J)$ [Eq. (66)] for the independent-rotor potential is consistently *smaller* than that for the corresponding coupled-rotor potential [$W_{\text{TM}}^{\text{CSHR}}(E, J)$, Eq. (67)]. The origin of this underestimate becomes apparent from a comparison of the plots of the two alternative model potentials in Fig. 8. In Fig. 8(a) is plotted the coupled-rotor potential of Eq. (65); in Fig. 8(b) that of the independent-rotor potential [Eq. (64)]. In both cases the barrier to rotation V_0 is assigned a value of unity in arbitrary units. It can be seen that the maximum value of the independent-rotor hindering potential, at $\theta'_1 = \theta'_2 = \pi/2$, is twice that of the coupled rotor potential. Stated simply, the independent-rotor potential allows one to break the partially formed bond twice: once by rotating fragment 1 to $\theta'_1 = \pi/2$, and then again by rotating fragment 2 to $\theta'_2 = \pi/2$. This additional hindrance is spurious if one considers only the bonding interaction potential $V_b(\mathbf{q})$. However, if at small fragment separations one sought to introduce a certain amount of "soft" steric hindrance at low energies [the high-energy "hard" steric hindrance is incorporated exactly into Eq. (61)], it is possible that the higher peak in the independent-rotor potential at configurations where the fragments are not favorably aligned for bonding might even prove useful. In any case, except at high tempera-

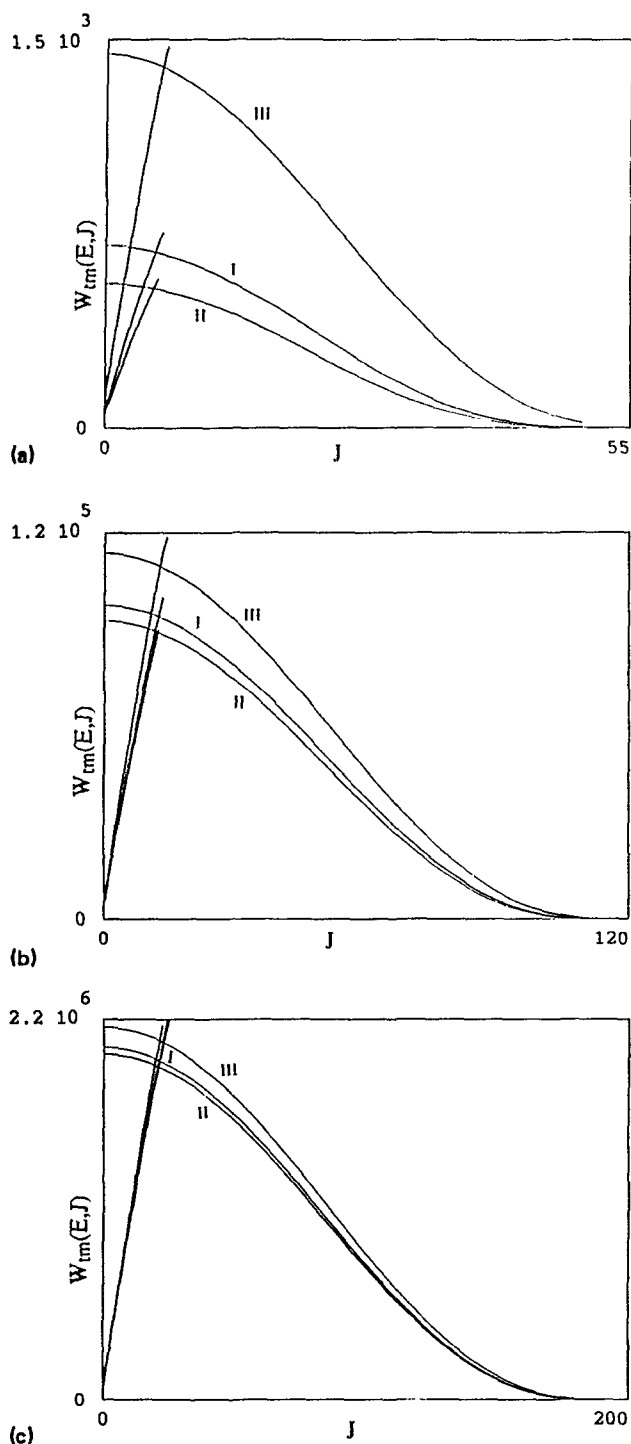


FIG. 7. Plot of transitional-mode sum of states $W_{TM}(E, J)$ vs angular momentum J for the $\text{CH}_3 + \text{CH}_3$ system, calculated for coupled sinusoidally hindered rotor potential (curve I), independent sinusoidally hindered rotor potential (curve II), and central potential (curve III). Center-of-mass separation 4.5 Å. Inertial parameters as for Fig. 6(a). Rotational barrier for hindered rotor potentials $V_0 = 100 \text{ cm}^{-1}$. (a) Available kinetic energy $\xi = 350 \text{ cm}^{-1}$. (b) Available kinetic energy $\xi = 1500 \text{ cm}^{-1}$. (c) Available kinetic energy $\xi = 4000 \text{ cm}^{-1}$.

tures, variationally selected transition states generally lie at larger fragment separations, where steric interactions are not significant (e.g., Ref. 16). Therefore, under most conditions a coupled-rotor form of the potential such as Eq. (65) is to be favored. Further considerations of this sort are spec-

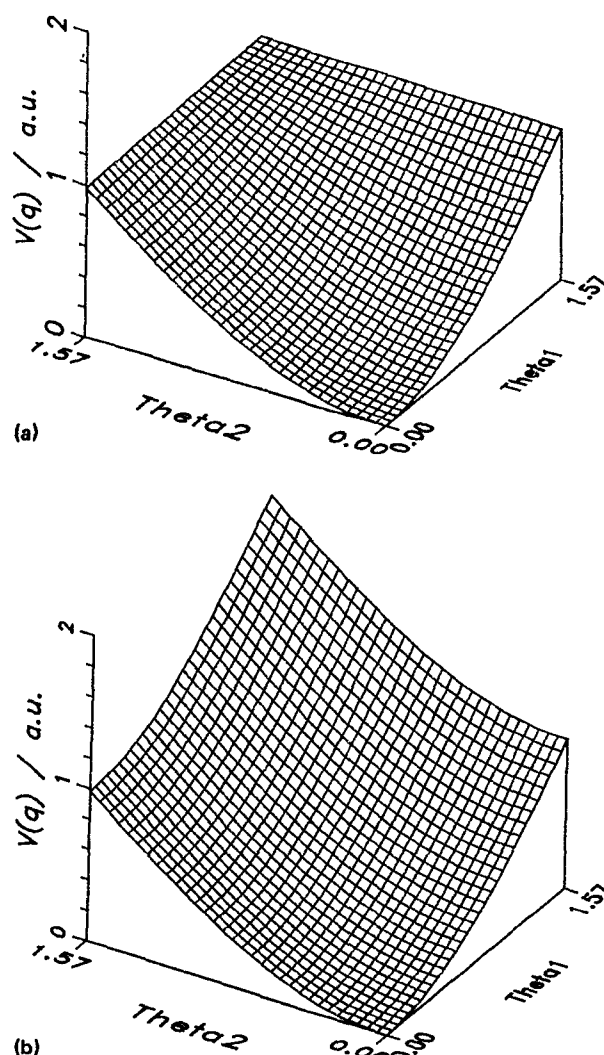


FIG. 8. Plot of sinusoidally hindered-rotor potentials for $V_0 = 1$ (arbitrary units). (a) Coupled sinusoidally hindered rotor potential [Eq. (65)]. (b) Independent sinusoidally hindered rotor potential [Eq. (64)].

ulative in the absence of more detailed information concerning the nature of the bonding and nonbonding interactions in the range of fragment separations of interest.

The formulas for systems containing fragments with asymmetric radical orbitals, such as OH, which form a bond in the $\theta'_1 = 0$ orientation but not at $\theta'_1 = \pi$, are derived by a simple extension of the considerations above. Consider first the case of a system with two asymmetric radicals. One can identify four quadrants in the (θ'_1, θ'_2) configuration space of Eq. (63). The first quadrant contains those configurations where the radical orbital of each fragment is in the hemisphere which faces the other fragment. In this region, the bonding potential may be well represented by a coupled-rotor functional form such as that given in Eq. (65) [Fig. 8(a)]. In the other three quadrants, one or the other or both of the radical orbitals on the fragments faces away from the partner fragment. For such configurations there is little likelihood of any significant bonding overlap, hence, to a first order of approximation, the potential $V_b(q)$ may be as-

signed the constant value of $V_0 = V(r = \infty) - V(r)$ (recall that V_b does not include any contributions to the potential from steric interactions of nonbonded atoms). In order to evaluate the integral over the nonbonding quadrants of the (θ'_1, θ'_2) configuration space, $V_b(\theta'_1, \theta'_2)$ in Eq. (63) is set equal to V_0 . The integral over each of the nonbonding quadrants then yields a result of the form of Eq. (67) where $H_b(\xi)$ is replaced by

$$H_{nb}(\xi) = (\xi - V_0)^m. \quad (70)$$

Hence, for the system comprised of two asymmetric radicals, one obtains

$$W_{TM}^{CSHR}(E, J) = S[\xi] + \frac{C}{\sigma_1 \sigma_2} [H_b(\xi) + 3H_{nb}(\xi)],$$

since three of the quadrants in the (θ'_1, θ'_2) space are nonbonding and only one is bonding. This result may be summarized concisely for the symmetric + symmetric, symmetric + asymmetric, and asymmetric + asymmetric cases as

$$W_{TM}^{CSHR}(E, J) = S[\xi] + \frac{C}{4\sigma_1 \sigma_2} [g_1 g_2 H_b(\xi) + (4 - g_1 g_2) H_{nb}(\xi)], \quad (71)$$

where

$$g_i = \begin{cases} 1 & \text{if fragment } i \text{ has an asymmetric radical orbital,} \\ 2 & \text{if fragment } i \text{ has a symmetric radical orbital,} \end{cases}$$

and $H_b(\xi)$ and $H_{nb}(\xi)$ are given, respectively, by Eqs. (68) and (70).

Comparison of Figs. 7(a)–7(c) shows that, as is to be expected, both the coupled-rotor and the independent-rotor sums of states approach the central potential (i.e., free rotor) limit at high energies, where $\xi \gg V_0$ and so the barrier to rotation is only a small perturbation on the total available energy. This free-rotor limit is given by Eq. (61) with $\Delta\Omega$ therein set to unity. However, recall that, if the fragments are sufficiently close together, a certain fraction of the configuration space is of necessity excluded due to nonbonded steric interactions (i.e., $\Delta\Omega \leq 1$). Under such conditions, the free-rotor limit attained by Eqs. (66) and (67) [i.e., Eq. (61) with $\Delta\Omega = 1$] will *overestimate* the correct high-energy-limiting value for $W_{TM}(E, J)$. Rather, the correct high-energy limit for $W_{TM}(E, J)$ will be given by Eq. (61) with the value of $\Delta\Omega$ calculated in Eq. (60) included. As noted above, since Eq. (61) includes no other hindrance except that due to the “hard” steric repulsion between the nonbonded atoms, it provides an *upper bound* to the actual sum of states. This upper-bound property is valid provided that the hard-sphere collision diameters d_{ij}^{HS} used to calculate $\Delta\Omega$ [Eq. (60)] are chosen as realistic *minimum* separations between the nonbonding atoms. Given the upper-bound property of Eq. (61), the sum of states $W_{TM}(E, J)$ should always be chosen as the minimum of that calculated on the basis of the model bonding potential [e.g., Eq. (66) and (67)] and that calculated on the basis of the hard steric hindrance [Eq. (61)]:

$$W_{TM}(E, J) = \min\{W_{TM}^b(E, J), W_{TM}^{HS}(E, J)\}. \quad (72)$$

Finally, we note that the approximate analytical results derived above assume that the dominant feature of the non-central potential is the chemical bonding, which introduces a well in the potential in one or more quadrants of the (θ'_1, θ'_2) plane. Other effects, such as the dependence of the bonding potential on the torsional angle ϕ_i and possible longer-range dipole–dipole interaction, are neglected. With the advantages in terms of simplicity provided by the present formulation, one could pursue additional analytical formulas in order to account for additional features on the potential surface. In order to avoid a profusion of analytic formulas, however, it would seem that such developments are best guided by accurate numerical solutions, where the use of a variety of more accurate models enable one to determine the kinetically important features of the potential surface.

V. CONCLUSION

In the present work, a new method has been presented for rapid energy- and angular-momentum-resolved classical state counting in flexible transition states. The method concerns itself with those degrees of freedom in the flexible transition state which correspond to product rotations and relative translations (excluding the reaction coordinate), commonly called the “transitional modes.” Since the sum and density of states for these degrees of freedom may accurately be evaluated by classical techniques, the method of approach is that of evaluating the classical phase-space volume integral subject to energy and angular momentum constraints. The adiabatic statistical theory^{3,21,22,20,31,34} has been primarily formulated and applied within the context of quantum-state elaboration, whereas recent developments in microcanonical variational RRKM theory^{2,4,23,37,39,40,45,50} have tended to use classical methods. Hence, the results of the present analysis lend themselves most immediately to use within the VRRKM formalism.

The general technique involves the consideration of the total classical phase-space integral in two parts. First, the available momentum-space volume $\Phi(E, J, \mathbf{q})$ for an arbitrary fixed configuration of the system is considered. Exact results for $\Phi(E, J, \mathbf{q})$ for a system comprised of two spherical-top fragments have been derived in Sec. II [Eq. (34)]. The technique for derivation of the spherical-top + spherical-top results is readily generalized to consider cases where asymmetric fragments are involved. In Sec. III a general equation for $\Phi(E, J, \mathbf{q})$ has been presented [Eq. (46)] which simply and accurately approximates the exact result for all relevant combinations of atom, linear, spherical-top, symmetric-top, and asymmetric-top fragments. The consistency of the method presented with the action-angle variable approach has been established in Sec. II and Appendix B by comparison with appropriately modified phase-space theory.

Following the derivation of simple and accurate expressions for the available momentum-space volume, the integration of $\Phi(E, J, \mathbf{q})$ over the configuration space is then considered. Accurate analytical approximations to the transitional-mode sum of states $W_{TM}(E, J)$ [Eqs. (66) and

(67)] have been derived in Sec. IV for common model potentials with the use of two approximations: (a) a simpler yet reliable low- J form for $\Phi(E, J, \mathbf{q})$ is used [Eqs. (38) and (49)], and (b) the configurational dependence of the external rotational constants $A_{\text{ext}}(\mathbf{q})$ and $B_{\text{ext}}(\mathbf{q})$ is neglected by assigning these quantities their values at the reaction-coordinate configuration [Eqs. (56) and (57)]. The final results enable extremely rapid yet reliable evaluation of the transitional-mode sum of states with complete account for conservation of the total angular momentum J and inclusion of the effects of an anharmonic, noncentral hindering potential to the fragment rotations.

The methods developed in the present work for analyzing sequentially the momentum-space and the configuration-space integrations for the classical sum of states provide a number of physical insights into the nature of the approximations involved in simpler models of angular momentum conservation, and the most important aspects of the exact numerical solution. The simple $J \approx L$ model for angular momentum conservation,²⁵ which treats the fragment rotations as internal and independent (e.g., Ref. 16), has been shown in Sec. II to significantly *overestimate* the momentum-space volume under two conditions: (a) at low- J values for all systems, since the proper limits on K , the projection of $\mathbf{J}$ onto the body-fixed z axis of the overall system, are neglected, and (b) at all values of J for systems involving heavy fragments, since the “internal” rotational constants for the body-fixed x - and y -axis rotations in the kinetic-energy expression correspond to moments of inertia which are not properly reduced. Since the body-fixed x - and y -axis rotations involve *three* rotors (i.e., two fragments and the orbital rotation), a suitable method of deriving a kinetic-energy expression with properly reduced internal angular momenta has been shown to involve the three-body “center-of-mass” and “skewed-axis” momentum transformations. These transformations, well known in the context of translational three-body motion, find a novel application in the present analysis.

The exact momentum-space volume has been shown to exhibit characteristic low- and high- J regimes (equivalent to those used by Troe in the context of phase-space theory²⁰), a fact which was used in deriving the approximate analytic results for $W_{\text{TM}}(E, J)$. The presence of a configurational dependence in the available momentum-space volume when the transition state contains asymmetric fragments [distinct from the configurational dependence introduced by the non-central potential $V(\mathbf{q})$] has been identified and clearly illustrated in Sec. III. Although this effect has been neglected in the derivation of analytic approximations to $W_{\text{TM}}(E, J)$ in Sec. IV, for reasons discussed therein, its significance for cases where it is most extreme (e.g., in reactions involving two linear fragments) may warrant further consideration in the future.

The effect on the resulting sum of states of using an independent-rotor functional form for the fragment interaction potential, as opposed to the more physically realistic coupled-rotor functional form, has been illustrated for the first time in the present work. In a comparison of the results for analogous coupled and independent sinusoidally hindered rotor functional forms, the independent-rotor func-

tional form has been shown to *underestimate* the sum of states. One can observe in conclusion that this underestimation due to the independent-rotor potential must to a certain extent offset the *overestimate* of the momentum-space volume due to the incorrect kinetic-energy expression used in the simple independent-rotor treatment.

The analysis of this work provides several improved levels of implementation of classical state counting in flexible transition states. The most accurate involves use of the general analytic expressions for $\Phi(E, J, \mathbf{q})$, followed by numerical evaluation of the configuration-space integral for $W_{\text{TM}}(E, J)$. This procedure involves at most a five-dimensional numerical integral, and may be expected to approximate excellently exact numerical results obtained at considerably greater effort by other methods.^{2,23} The analytic expressions obtained in Sec. IV for $W_{\text{TM}}(E, J)$ represent a substantial improvement on the more common and simpler RRKM models (e.g., Ref. 16), and may also be expected to approximate exact numerical results accurately.

Finally, we note that an additional consideration in the application of variational RRKM theory (and variational transition-state theory in general) is the selection of an appropriate reaction coordinate. The reaction coordinate used in the present study is the center-of-mass separation between the fragments. Klippenstein⁵³ has recently introduced an implementation of variational RRKM theory which uses instead the separation between the two bonding atoms, which are often not located at the centers of mass of the fragments, and shown that this can lead to a reduction in the predicted rate coefficient.

A major factor which has hindered the use of variational RRKM theory on a routine basis in the past has been the computational complexity involved in evaluating the sum of states, successively over a range of values of the reaction coordinate, for the entire range of E and J values required in the calculation of a thermal rate coefficient. It is to be expected that the developments of this work will contribute substantially to the overcoming of this problem and facilitate the use of variational RRKM theory on a routine basis in the future for the accurate prediction of rate coefficients in a wide range of chemical reactions.

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APPENDIX A: PROOF OF EXPRESSION FOR EXTERNAL ROTATIONAL CONSTANTS

In this appendix, a simplified expression is derived for the moment of inertia with respect to external rotation of a flexible transition state about an arbitrarily oriented axis passing through the overall center of mass. This proves the

identity established in Eq. (27a) concerning the external principal-axis rotational constants of a flexible transition state consisting of two spherical-top fragments.

Consider a system composed of two collections of mass points, corresponding to the flexible transition state with two molecular fragments. Let $\{m_1, \dots, m_n\}$ be the masses in the first group, with total mass $M_1 = \sum_i m_i$ ($i = 1, \dots, n$), and $\{m_{n+1}, \dots, m_N\}$ be the masses in the second group, with total mass $M_2 = \sum_j m_j$ ($j = n + 1, \dots, N$). Let the origin be located at the center of mass of the entire system. The position vector $\mathbf{R}_i$ of the center of mass of each group is defined in the usual manner:

$$\mathbf{R}_i = M_i^{-1} \sum m_i \mathbf{r}_i, \quad (\text{A1})$$

where $\mathbf{r}_i$ is the position vector of the point i relative to the center of mass of the entire system. Let $\mathbf{r}'_i$ be the position vector of point i relative to the center of mass of the group to which it belongs, then

$$\sum_i m_i \mathbf{r}'_i \equiv 0. \quad (\text{A2})$$

Let $\mathbf{n}$ be a unit vector in the direction of the axis of rotation. The moment of inertia I for rotation of the overall system about the given axis is

$$\begin{aligned} I &= \sum_{i=1}^N m_i (\mathbf{r}_i \times \mathbf{n}) \cdot (\mathbf{r}_i \times \mathbf{n}) \\ &= \sum_{i=1}^n m_i (\mathbf{r}_i \times \mathbf{n}) \cdot (\mathbf{r}_i \times \mathbf{n}) \\ &\quad + \sum_{j=n+1}^N m_j (\mathbf{r}_j \times \mathbf{n}) \cdot (\mathbf{r}_j \times \mathbf{n}), \end{aligned} \quad (\text{A3})$$

where $(\mathbf{r}_i \times \mathbf{n}) \cdot (\mathbf{r}_i \times \mathbf{n})$ is the square of the perpendicular distance from the rotation axis to the point i . Examining the first of the summations on the right-hand side of Eq. (A3), one has

$$\begin{aligned} \sum_{i=1}^n m_i (\mathbf{r}_i \times \mathbf{n}) \cdot (\mathbf{r}_i \times \mathbf{n}) &= \sum_{i=1}^n m_i [(\mathbf{R}_1 + \mathbf{r}'_i) \times \mathbf{n}] \cdot [(\mathbf{R}_1 + \mathbf{r}'_i) \times \mathbf{n}] \\ &= \sum_{i=1}^n m_i [(\mathbf{R}_1 \times \mathbf{n}) \cdot (\mathbf{R}_1 \times \mathbf{n}) + 2(\mathbf{R}_1 \times \mathbf{n}) \cdot (\mathbf{r}'_i \times \mathbf{n}) \\ &\quad + (\mathbf{r}'_i \times \mathbf{n}) \cdot (\mathbf{r}'_i \times \mathbf{n})] \\ &= M_1 (\mathbf{R}_1 \times \mathbf{n}) \cdot (\mathbf{R}_1 \times \mathbf{n}) \\ &\quad - 2(\mathbf{R}_1 \times \mathbf{n}) \cdot \left(\mathbf{n} \times \sum_{i=1}^n m_i \mathbf{r}'_i \right) + I_1, \end{aligned} \quad (\text{A4})$$

where I_1 is the moment of inertia of the first group with respect to rotation about an axis passing through the center of mass of group 1 which is parallel to that under consideration. The middle term of Eq. (A4) is zero due to the fundamental definition of Eq. (A2). Hence Eq. (3) may be written as

$$\begin{aligned} I &= M_1 (\mathbf{R}_1 \times \mathbf{n}) \cdot (\mathbf{R}_1 \times \mathbf{n}) \\ &\quad + M_2 (\mathbf{R}_2 \times \mathbf{n}) \cdot (\mathbf{R}_2 \times \mathbf{n}) + I_1 + I_2 \\ &= \mu R^2 + I_1 + I_2, \end{aligned} \quad (\text{A5})$$

where μ is the reduced mass of the two groups and R the separation between their centers of mass as viewed along the axis of rotation.

Equation (A5) may now be applied to the case of the flexible transition state with two spherical-top fragments whose centers of mass are separated by a distance r . The overall transition state is a prolate top, and hence there are two principal axes perpendicular to the line joining the centers of mass with degenerate moments of inertia, the third principal axis with the unique (smallest) moment of inertia I_A lying along the axis joining the centers of mass. The separation between the centers of mass as viewed along the symmetry axis is zero, hence $I_A = I_1 + I_2$. The separation as viewed along an axis perpendicular to the line joining the centers of mass is, of course, r ; hence $I_B = \mu r^2 + I_1 + I_2$. The rotational constants follow directly as in Eqs. (27a) and (27b).

APPENDIX B: DERIVATION OF SPHERICAL-TOP-SPHERICAL-TOP RESULTS VIA ACTION-ANGLE VARIABLES

In PST, the classical phase-space integral for the sum of states is written in terms of the action variables [e.g., Ref. 30(e)]:

$$\frac{W_{\text{PST}}(E, J)}{(2J + 1)} = \frac{1}{\sigma_1 \sigma_2} \int \cdots \int dL dj_r dK_1 dj_2 dK_2 \times \Delta(J, j_r, L) \Delta(j_r, j_1, j_2) S[E^* - H_{\text{TM}}], \quad (\text{B1})$$

where E is the total energy, J is the total angular momentum, L is the orbital angular momentum, j_r is the total angular momentum of the fragment pair, j_i is the total angular momentum of fragment i , and K_i is the body-fixed z -axis component of j_i [i.e., $K_i = j'_{iz}$, where j'_{iz} is defined as in Eqs. (7), (8), and (9)]. The appropriate angular momentum vector relations are

$$\mathbf{j}_r = \mathbf{j}_1 + \mathbf{j}_2 \quad (\text{B2})$$

and

$$\mathbf{J} = \mathbf{L} + \mathbf{j}_r. \quad (\text{B3})$$

The limits on the allowed magnitudes of the angular momenta that arise from Eqs. (B2) and (B3) are summarized in Eq. (B1) by the triangle inequalities:

$$\begin{aligned} \Delta(J, j_r, L) &= \begin{cases} 1, & |j_r - L| \leq J \leq |j_r + L| \\ 0, & \text{otherwise} \end{cases}, \\ \Delta(j_r, j_1, j_2) &= \begin{cases} 1, & |j_1 - j_2| \leq j_r \leq |j_1 + j_2| \\ 0, & \text{otherwise} \end{cases}. \end{aligned}$$

In addition, the values of K_1 and K_2 in Eq. (B1) are restricted such that $|K_1| \leq j_1$ and $|K_2| \leq j_2$. The Hamiltonian H_{TM} is that of Eq. (6) with $V(\mathbf{q})$ therein set to zero; in terms of the present action variables it takes the simple form:

$$H_{\text{TM}} = B_0 L^2 + B_1 j_1 + B_2 j_2^2. \quad (\text{B4})$$

The step function $S[E^* - H_{\text{TM}}]$, where $E^* = E - V(r)$, ensures that only states with total energy less than or equal to E are counted. The factor of $(2J + 1)$ in Eq. (B1) allows for

the M degeneracy of the total angular momentum vector $\mathbf{J}$, where M is the component of $\mathbf{J}$ with respect to an arbitrary spatially fixed axis [the integral on the right-hand side of Eq. (B1), which evaluates the available phase-space volume in units of h , is classical and will require appropriate "requantizing" after evaluation by setting $2J \rightarrow (2J + 1)$, $BJ^2 \rightarrow (BJ(J + 1))$. In Eq. (B1), the total angular degeneracy of $(2\pi)^8$ arising from integration over the angle variables is implicitly included (i.e., the angular integrations have already been carried out). The canonical action-angle variables may be derived from the conventional Euler angles and their conjugate momenta by canonical transformation (e.g., Ref. 38).

In order to proceed with the evaluation of Eq. (B1), the sum of states $W_r(E_{\text{rmax}} j_r, m_r)$ is first defined, which represents the total phase-space volume available to the fragment pair for an angular momentum j_r , projection onto an arbitrary spatially fixed axis m_r , and maximum energy E_{rmax} :

$$W_r(E_{\text{rmax}}, j_r, m_r) = \frac{1}{\sigma_1 \sigma_2} \int \cdots \int dj_1 dj_2 dK_1 dK_2 \times \Delta(j_r, j_1, j_2) S[E_{\text{rmax}} - H_f], \quad (\text{B5})$$

where H_f is the free-rotational Hamiltonian for the fragments:

$$H_f = B_1 j_1^2 + B_2 j_2^2. \quad (\text{B6})$$

$W_r(E_{\text{rmax}}, j_r, m_r)$ may be evaluated by standard methods^{30(c)} to give

$$W_r(E_{\text{rmax}}, j_r, m_r) = \frac{1}{\sigma_1 \sigma_2} \frac{8j_r}{3} \frac{1}{(B_1 + B_2)^{3/2}} \times (E_{\text{rmax}} - B_{12} j_r^2)^{3/2}, \quad (\text{B7})$$

so that $W_{\text{PST}}(E, J)$ is given by

$$\frac{W_{\text{PST}}(E, J)}{(2J + 1)} = \frac{1}{\sigma_1 \sigma_2} \frac{8}{3(B_1 + B_2)^{3/2}} \int dL \int dj_r \times j_r (E_{\text{rmax}} - B_{12} j_r^2)^{3/2} \Delta(J j_r, L). \quad (\text{B8})$$

The value of E_{rmax} in the case of a central potential, wherein L is conserved, is determined in PST by the criterion of locating the orbiting transition state at the position of the centrifugal barrier generated by the effective potential $V_{\text{eff}}(L, r)$ (e.g., Ref. 28):

$$V_{\text{eff}}(L, r) = B_0(r) L^2 + V(r), \quad (\text{B9})$$

where $B_0(r) = h^2/8\pi^2\mu r^2$, μ being the reduced mass of the two fragments. The present purpose, however, is to use the central-potential limit to establish the consistency of results generated by the action-angle variable approach with the results of this work when applied under the same circumstances. Since the present work addresses itself to the case of a noncentral potential, wherein L is *not* conserved, it is necessary to drop the orbiting-transition-state criterion from PST in order to have a proper comparison. One therefore locates the transition state at a fixed separation r for *all* orbital angular momenta L , and counts all states which are energetically accessible at the given separation. E_{rmax} is then given by

$$E_{\text{rmax}} = E^* - B_0 L^2, \quad (\text{B10})$$

where B_0 is as defined above. Consistent with Eq. (40), we define $W'_{\text{CP}}(E, J)$ as

$$\frac{W'_{\text{CP}}(E, J)}{(2J + 1)} = \frac{1}{\sigma_1 \sigma_2} \frac{8}{3(B_1 + B_2)^{3/2}} \int dL \int dj_r \times (E^* - B_0 L^2 - B_{12} j_r^2)^{3/2} \Delta(J j_r, L). \quad (\text{B11})$$

Our objective is to show that $W'_{\text{CP}}(E, J)$, obtained via the action-angle variable approach, is indeed the same as $W_{\text{CP}}(E, J)$ of Eq. (40).

Figure 4 illustrates the boundaries to the integral in the (j_r, L) plane imposed by the $\Delta(J j_r, L)$ constraint and the maximum energy E^* . The shaded region (I) is the area over which the integral must be evaluated. The integration boundaries may be simplified as follows. Let Y_i represent the integral of the integrand in Eq. (B11) over region i as indicated in Fig. 4. Noting first that the integrand is odd with respect to j_r , one has

$$Y_{\text{II}} = -Y_{\text{III}}. \quad (\text{B12})$$

Noting that the integrand is even with respect to L , one has

$$Y_{\text{IV}} = Y_{\text{V}}. \quad (\text{B13})$$

Let $Y_\alpha = Y_{\text{I}} + Y_{\text{II}} + Y_{\text{III}} + Y_{\text{IV}}$, and $Y_\beta = Y_\alpha + Y_{\text{V}}$. then,

$$\frac{W'_{\text{CP}}(E, J)}{(2J + 1)} = Y_{\text{I}} = 2Y_\alpha - Y_\beta, \quad 0 \leq J \leq J^*, \quad (\text{B14})$$

where $J^* = (E^*/A_{\text{ext}})^{1/2}$, with $A_{\text{ext}} = B_{12}$ for the sphere-sphere system. For higher J values, regions IV and V of Fig. 4 disappear, and the boundaries alter such that

$$\frac{W'_{\text{CP}}(E, J)}{(2J + 1)} = Y_\beta, \quad J^* \leq J \leq J_{\text{max}}, \quad (\text{B15})$$

where $J_{\text{max}} = (E^*/B_{\text{ext}})^{1/2}$, with $B_{\text{ext}} = B_{012}$ for the sphere-sphere system. Y_α is written as

$$Y_\alpha = \frac{1}{\sigma_1 \sigma_2} \frac{8}{3(B_1 + B_2)^{3/2}} \times \int_0^{L_{\text{max}}} dL \int_{J-L}^{j_{\text{rmax}}} dj_r j_r (E^* - B_0 L^2 - B_{12} j_r^2)^{3/2}, \quad (\text{B16})$$

where

$$L_{\text{max}} = \frac{1}{(B_0 + B_{12})} \{B_{12} J + [(B_0 + B_{12}) \times (E^* - B_{012} J^2)]^{1/2}\} \quad (\text{B17})$$

and

$$j_{\text{rmax}} = \left[\frac{(E^* - B_0 L^2)}{B_{12}} \right]^{1/2}. \quad (\text{B18})$$

Evaluation of Eq. (B16), followed by appropriate requantizing of J , yields the following result:

$$Y_\alpha = \frac{1}{2\sigma_1 \sigma_2} [\Phi(E, \text{low } J) + \Phi(E, \text{high } J)], \quad (\text{B19})$$

where $\Phi(E, \text{low } J)$ is given by Eq. (34a) with $V(\mathbf{q})$ therein set to zero, and $\Phi(E, \text{high } J)$ is given by Eq. (34b) with $V(\mathbf{q})$ also set to zero. Y_β is written as

$$Y_\beta = \frac{1}{\sigma_1 \sigma_2} \frac{8}{3(B_1 + B_2)^{3/2}} \times \int_{L_{\min}}^{L_{\max}} dL \int_{J-L}^{J_{\max}} dj_r (E^* - B_0 L^2 - B_{12} j_r^2)^{3/2}, \quad (\text{B20})$$

where

$$L_{\min} = \frac{1}{(B_0 + B_{12})} \{B_{12} J - [(B_0 + B_{12}) \times (E^* - B_{012} J^2)]^{1/2}\}. \quad (\text{B21})$$

Evaluation of Eq. (B20) followed by requantization of J yields

$$Y_\beta = \Phi(E, \text{high } J) / \sigma_1 \sigma_2. \quad (\text{B22})$$

Equations (B14), (B15), (B19), and (B22) together yield the desired result of Eq. (40). This establishes consistency between the method of the present paper and the action-angle variable method.

APPENDIX C: THE TRANSFORMATION FROM SPATIALLY REFERENCED TO BODILY REFERENCED FRAGMENT EULER ANGLES

In this appendix, the transformation from “absolute” fragment Euler angles, which are referenced to the spatially

fixed Cartesian axes, to “relative” Euler angles, which are referenced to the orbital principal axes, is considered.

The principal axial system (x, y, z) of the orbital rotor is related to the spatially fixed axial system (x_s, y_s, z_s) by the orbital Euler angles (θ, ϕ) through the rotation matrix A_0 , such that the projection $\mathbf{v}$ in the orbital system of a vector $\mathbf{v}_s$ in the spatially fixed system is given by

$$\mathbf{v} = A_0 \mathbf{v}_s, \quad (\text{C1})$$

where A_0 is the rotation matrix (e.g., Ref. 46)

$$A_0 = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi \cos \theta & \cos \phi \cos \theta & \sin \theta \\ \sin \phi \sin \theta & -\cos \phi \sin \theta & \cos \theta \end{bmatrix}. \quad (\text{C2})$$

The principal axial system (x'_i, y'_i, z'_i) of the fragment i is related similarly to the spatially fixed Cartesian axial system by the absolute fragment Euler angles $(\theta_i, \phi_i, \psi_i)$ through the rotation matrix A such that

$$\mathbf{v}' = A \mathbf{v}_s, \quad (\text{C3})$$

where A is given by

$$A = \begin{bmatrix} \cos \psi_i \cos \phi_i - \cos \theta_i \sin \phi_i \sin \psi_i & -\sin \psi_i \cos \phi_i - \cos \theta_i \sin \phi_i \cos \psi_i & \sin \theta_i \sin \phi_i \\ \cos \psi_i \sin \phi_i + \cos \theta_i \cos \phi_i \sin \psi_i & -\sin \psi_i \sin \phi_i + \cos \theta_i \cos \phi_i \cos \psi_i & -\sin \theta_i \cos \phi_i \\ \sin \psi_i \sin \theta_i & \cos \psi_i \sin \theta_i & \cos \theta_i \end{bmatrix}. \quad (\text{C4})$$

We wish to determine the relative Euler angles $(\theta'_i, \phi'_i, \psi'_i)$ that would be required to rotate the fragment, starting from a configuration where its principal axes (x'_i, y'_i, z'_i) are co-aligned with the orbital principal axes (x, y, z) into the configuration specified by the absolute (or, spatially referenced) fragment Euler angles $(\theta_i, \phi_i, \psi_i)$.

First, the projection $\mathbf{u}$ onto the orbital principal axial system (x, y, z) of a unit vector $\mathbf{u}'_z$ oriented along the fragment z'_i principal axis must be determined. The rotation matrices A and A_0 furnish the required relationship:

$$\mathbf{u} = A_0 A^{-1} \mathbf{u}'_z, \quad (\text{C5})$$

since $\mathbf{u}'_z$ is the vector $(0, 0, 1)$, we have

$$\mathbf{u} = \begin{pmatrix} a_{13} \\ a_{23} \\ a_{33} \end{pmatrix}, \quad (\text{C6})$$

where a_{13} , a_{23} , and a_{33} are the elements of the right-hand column of the matrix product $A_0 A^{-1}$ (note that, since the rotation matrices are orthogonal, $A^{-1} = A^T$).⁴⁶

$$a_{13} = \cos \phi \sin \phi_i \sin \theta_i - \sin \phi \cos \phi_i \sin \theta_i, \quad (\text{C7})$$

$$a_{23} = -\sin \phi \cos \theta \sin \phi_i \sin \theta_i - \cos \phi \cos \theta \cos \phi_i \sin \theta_i + \sin \theta \cos \theta_i, \quad (\text{C8})$$

$$a_{33} = \sin \phi \sin \theta \sin \phi_i \sin \theta_i + \cos \phi \sin \theta \cos \phi_i \sin \theta_i + \cos \theta \cos \theta_i. \quad (\text{C9})$$

In the rotation of the fragment from the configuration specified by (θ, ϕ) , which is the new reference configuration for the fragment Euler angles, to the configuration specified by $(\theta_i, \phi_i, \psi_i)$, the z principal axis of the fragment moves from being aligned with the orbital z axis to being aligned in the direction of $\mathbf{u}$. Hence the relative Euler angle θ'_i is determined by the scalar product of $\mathbf{u}$ and a unit vector along the orbital z axis:

$$\begin{aligned} \cos \theta'_i &= \begin{pmatrix} a_{13} \\ a_{23} \\ a_{33} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ &= \sin \phi \sin \theta \sin \phi_i \sin \theta_i \\ &\quad + \cos \phi \sin \theta \cos \phi_i \sin \theta_i + \cos \theta \cos \theta_i. \end{aligned} \quad (\text{C10})$$

The relative Euler angle ϕ'_i is determined from the projection of $\mathbf{u}$ onto the (x, y) plane of the orbital principal axial system. One has

$$\tan \phi'_i = \frac{-a_{13}}{a_{23}} = \frac{\sin \phi \cos \phi_i \sin \theta_i - \cos \phi \sin \phi_i \sin \theta_i}{-\sin \phi \cos \theta \sin \phi_i \sin \theta_i - \cos \phi \cos \theta \cos \phi_i \sin \theta_i + \sin \theta \cos \theta_i}. \quad (\text{C11})$$

Finally, ensuring that the total rotation about the fragment z principal axis is the same when rotating directly through $(\theta_i, \phi_i, \psi_i)$ as when rotating through (θ, ϕ) followed by $(\theta'_i, \phi'_i, \psi'_i)$, one obtains

$$\psi'_i = \phi_i + \psi_i - \phi - \phi'_i. \quad (\text{C12})$$

Substituting for ϕ'_i from Eq. (C11) gives for ψ'_i

$$\psi'_i = \phi_i + \psi_i - \phi - \tan^{-1} \left(\frac{\sin \phi \cos \phi_i \sin \theta_i - \cos \phi \sin \phi_i \sin \theta_i}{-\sin \phi \cos \theta \sin \phi_i \sin \theta_i - \cos \phi \cos \theta \cos \phi_i \sin \theta_i + \sin \theta \cos \theta_i} \right). \quad (\text{C13})$$

Equations (C10), (C11), and (C13) specify the necessary transformation equations from the absolute Euler coordinates $(\theta_i, \phi_i, \psi_i)$ to the relative Euler coordinates $(\theta'_i, \phi'_i, \psi'_i)$. The Jacobian for the transformation may be determined at some length by the usual method. However, a simpler method suffices as shown below. Consider the integral

$$I = \int d\psi_i \int d\phi_i \int d\theta_i \sin \theta_i f(\phi_i, \theta_i, \psi_i). \quad (\text{C14})$$

First, introduce the dummy-variable r_i :

$$I = \int d\psi_i \int d\phi_i \int d\theta_i \sin \theta_i \times \int dr_i r_i^2 f(\theta_i, \phi_i, \psi_i) \delta[r_i^2 - 1]. \quad (\text{C15})$$

Now transform (r_i, θ_i, ϕ_i) to the Cartesian coordinates (x_s, y_s, z_s) in the manner of a spherical-polar to Cartesian transformation, retaining ψ_i for the moment. The Jacobian for the spherical-polar to Cartesian transformation is $(r_i^2 \sin \theta_i)^{-1}$, so that

$$I = \int d\psi_i \int dx_s \int dy_s \int dz_s f(\psi_i, x_s, y_s, z_s) \times \delta[x_s^2 + y_s^2 + z_s^2 - 1]. \quad (\text{C16})$$

Now carry out a rotation of Cartesian axes from (x_s, y_s, z_s) to (x, y, z) . The Jacobian for this transformation is unity, and I becomes

$$I = \int d\psi_i \int dx \int dy \int dz f(\psi_i, x, y, z) \times \delta[(x^2 + y^2 + z^2) - 1]. \quad (\text{C17})$$

Transforming back to spherical-polar coordinates $(r'_i, \theta'_i, \phi'_i)$, which are now referenced to the (x, y, z) orbital-principal-axial system, one has

$$I = \int d\psi_i \int d\phi'_i \int d\theta'_i \sin \theta'_i \int dr'_i r_i'^2 \times f(\theta'_i, \phi'_i, \psi_i) \delta[r_i'^2 - 1]. \quad (\text{C18})$$

Collapsing the integral over r'_i , and finally transforming with unit Jacobian from ψ_i to ψ'_i using Eq. (C12). I becomes

$$I = \int d\psi'_i \int d\phi'_i \int d\theta'_i \sin \theta'_i f(\phi'_i, \theta'_i, \psi'_i). \quad (\text{C19})$$

Comparison of Eqs. (C14) and (C19) gives the desired Jacobian for the transformation, as summarized in Eq. (50).

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